Marine Microbial Biodiversity

Lecture 12

Marine Ecosystems Part 1

Matteo Selci, PhD

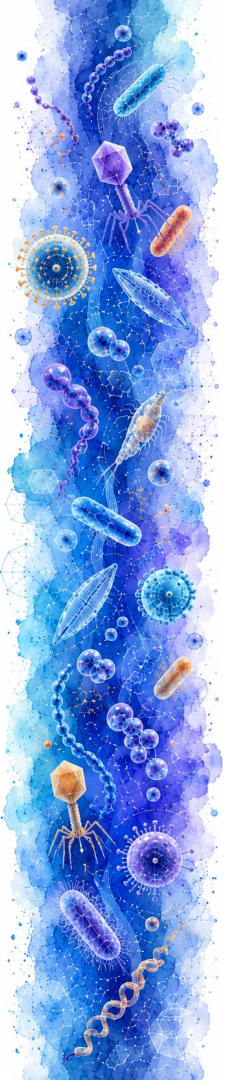
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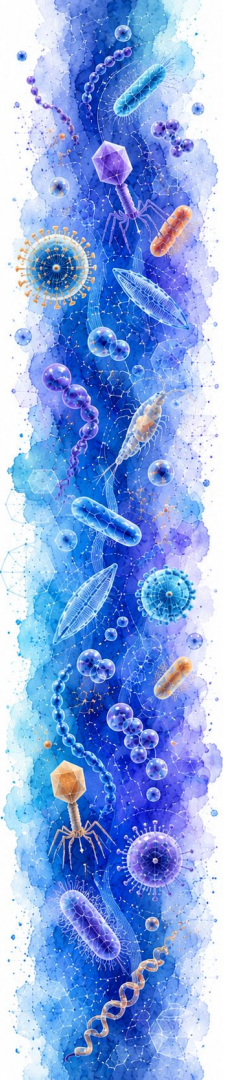
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Which are the limits of life?



Extremophiles Define The Boundary Conditions Of Life On Earth

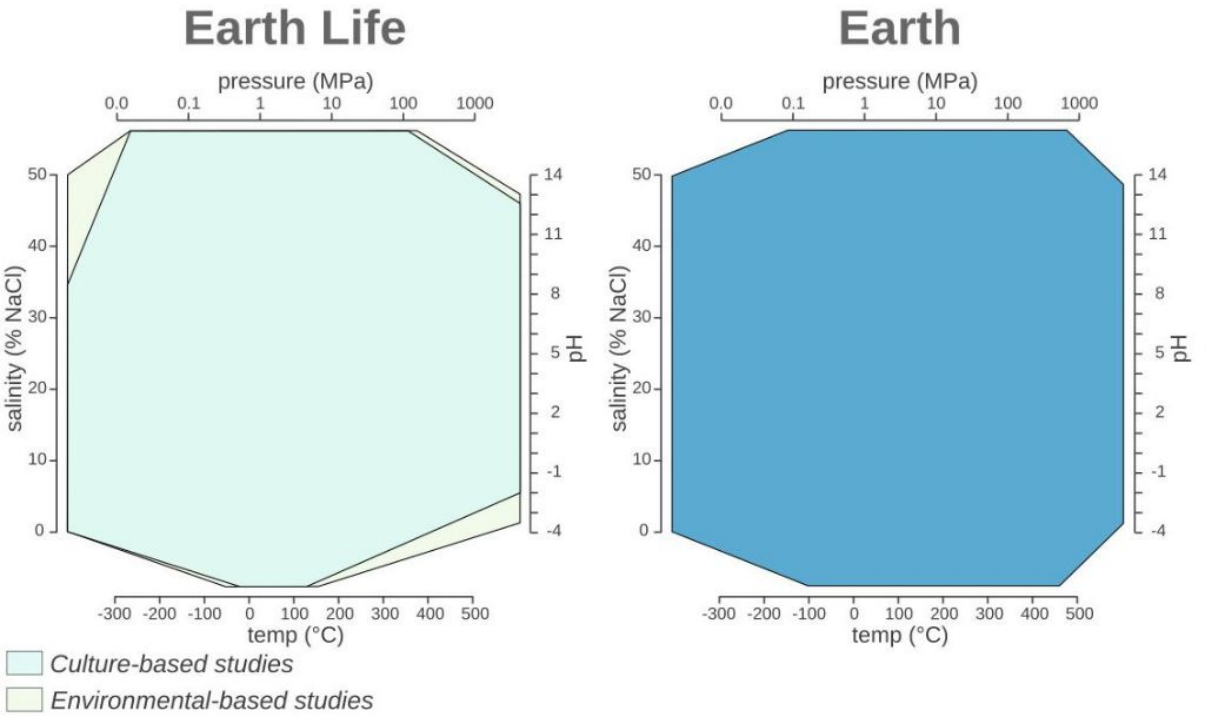
Strain	Domain	Extremophile Type	Isolation ecosystem	Temperature (°C)	pH	Pressure (Mpa)	Salinity (%)	Water activity (a_w)
Picrophilus oshimae KAW 2/2	*Archaea*	Hypercidophile	Hot springs, Solfataras	47–65 (60)[a]	**−0.06**–1.8 (0.7)	nr	0–20	nr
Serpentinomonas sp. B1	*Bacteria*	Alkaliphile	Serpentinizing system (water)	18–37 (30)	9–**12.5** (11)	nr	0–0.5 (0)	nr
Methanopyrus kandleri 116	*Archaea*	Hyperthermophile	Deep-sea hydrothermal vent	90–**122** (105)	(6.3–6.6)	0.4–40	0.5–4.5 (3.0)	nr
Planococcus halocryophilus Or1	*Bacteria*	Halopsychrophile	Sea ice core	**−15**–37 (25)	6–11 (7–8)	nr	0–19 (2)	nr
Halarsenatibacter silvermanii SLAS-1	*Bacteria*	Haloalkaliphile	Soda lake	28–55 (44)	8.7–9.8 (9.4)	nr	20–35 (**35**)	nr
Thermococcus piezophilus CDGS	*Archaea*	Piezothermophile	Deep-sea hydrothermal vent	60–95 (75)	5.5–9 (6)	0.1–**125** (50)	2–6 (3)	nr
Haloarchaeal strains GN-2 and GN-5	*Archaea*	Xerophile	Solar salterns (brine)	nr	nr	nr	nr	**0.635**

[a]*Data presented as range (optimum) for each parameter. nr, not reported in the original publication. Current limits are highlighted in bold.*

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10.3389/fmicb.2019.00780

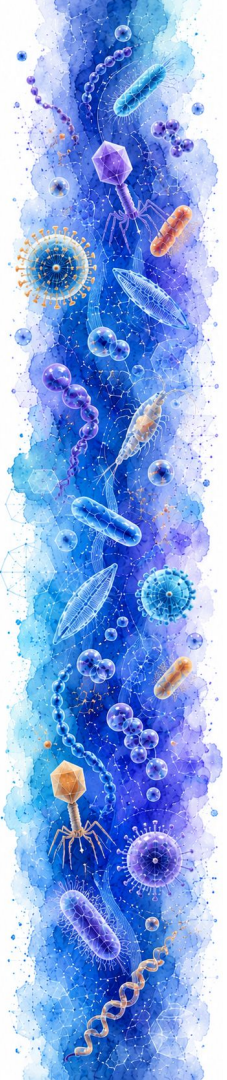
Extremophiles Define The Boundary Conditions Of Life On Earth



Life has occupied almost every possible niche on our planet, with few exception

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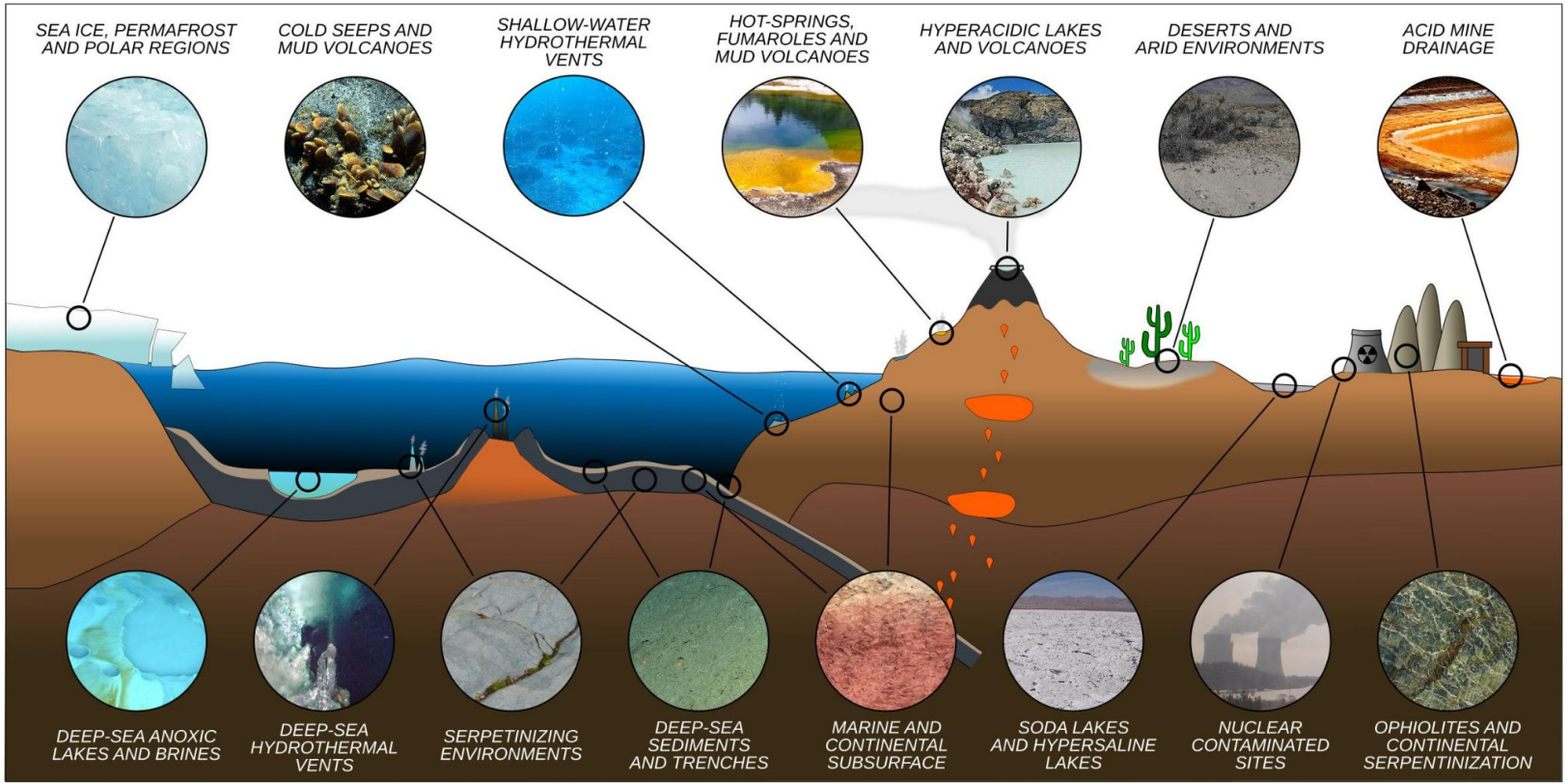
Extremophiles Define The Boundary Conditions Of Life On Earth

Biome	Temperature (°C)	pH	Pressure (MPa)	Salinity (% NaCl)
Deep-sea floor and trenches	−1.9–13.8[d]	7.3–8.1	2.1–112	3.4–3.9
Deep-sea hydrothermal vents	<1[e]–464	4–11	2.1–50.7	0.1–8
Deep hypersaline anoxic basins	10–65	5.4–8.6	2.1–40.5	4–50[f]
Subsurface ecosystems	3.25–<400[g]	~1–12.8	<800[h]	0.05–saturation[i]

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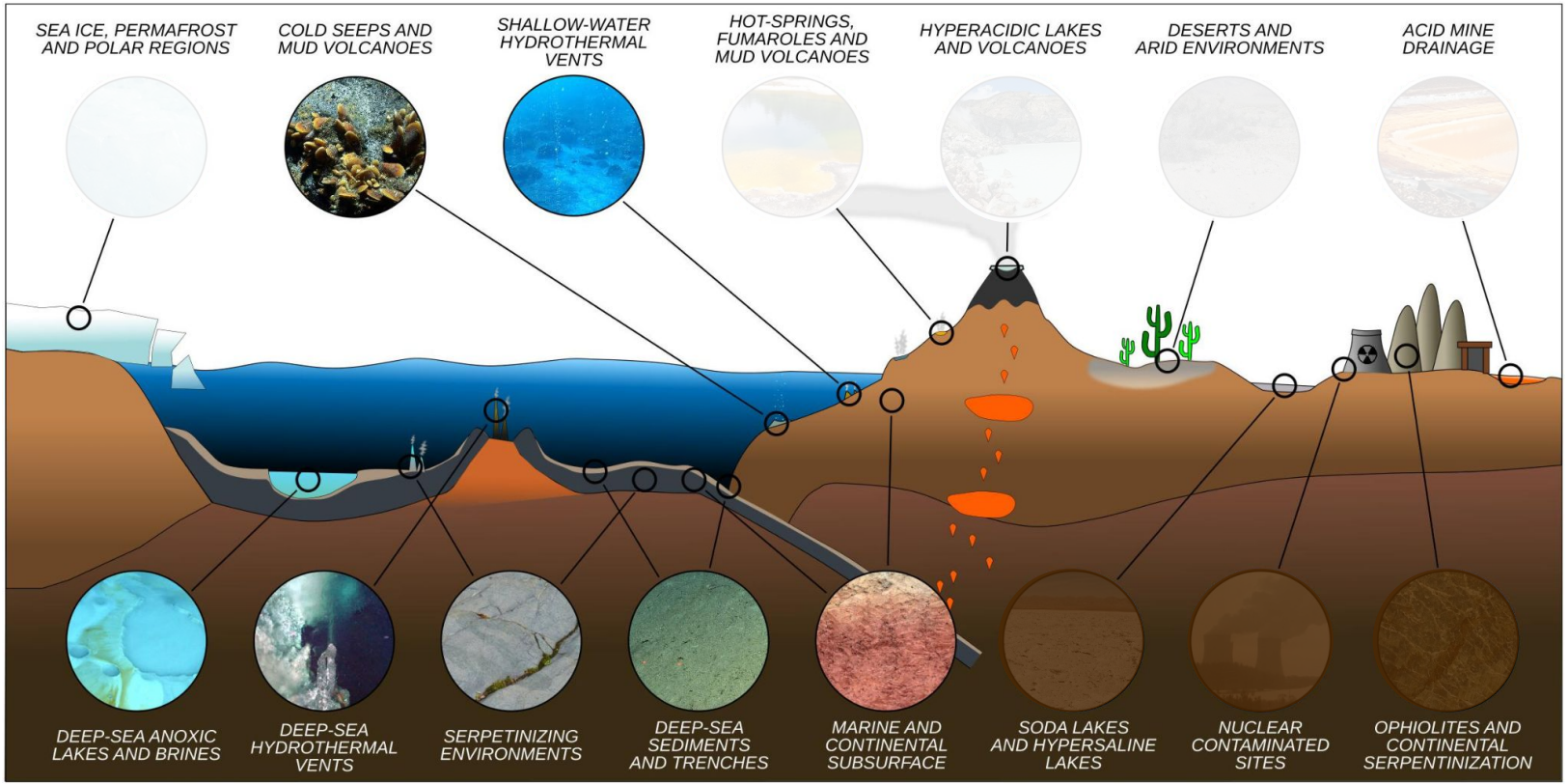
Diversity Of Ecosystems On Planet Earth



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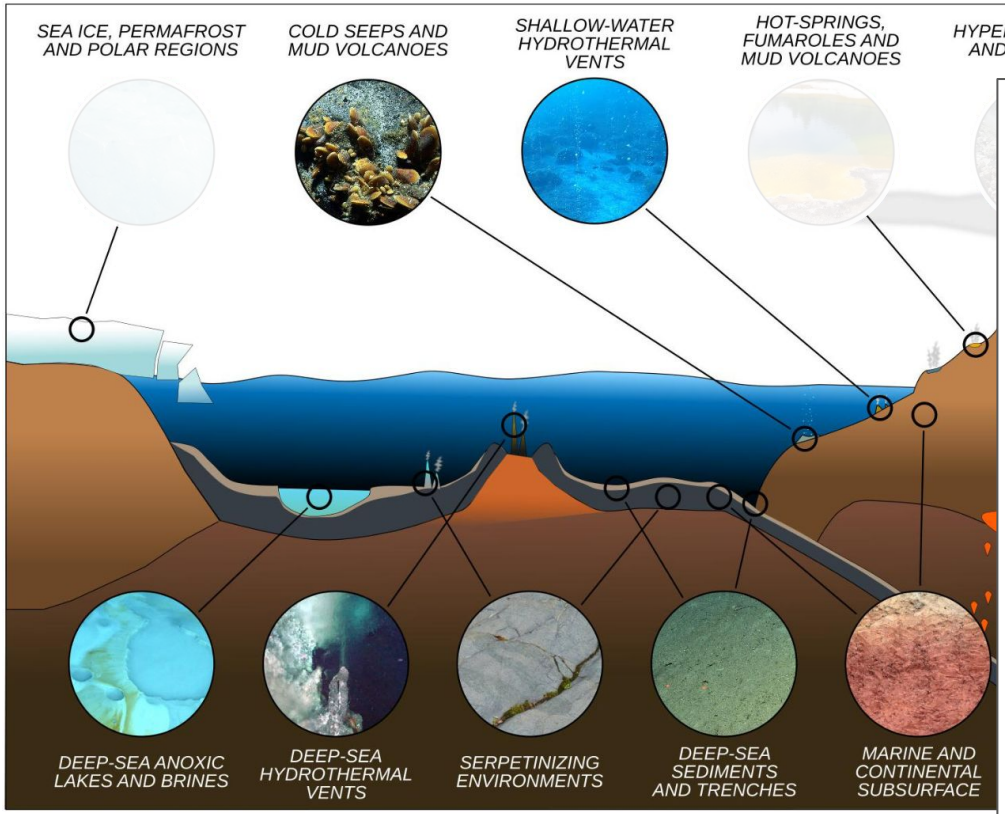
Diversity Of Ecosystems On Planet Earth



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Diversity Of Ecosystems On Planet Earth

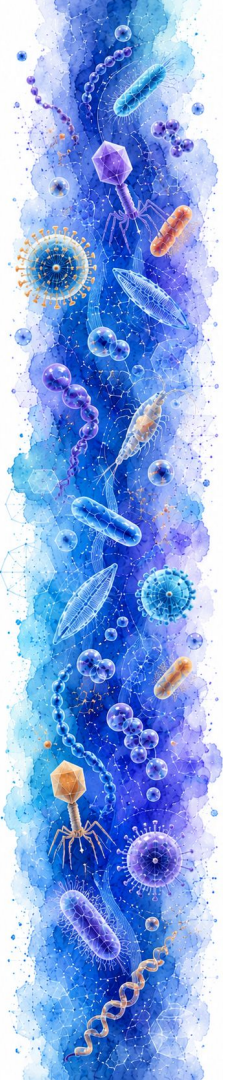


Marine ecosystems also include open ocean surface waters and sea floor, coastal zones, intertidal zones, sandy shores, rocky shores, mudflats, swamps and salt marshes, estuaries, kelp forests, coral reefs, brackish environments, seagrasses.

These ecosystems are "mainly" dominated by macro-/mega-fauna and, with few exceptions, we have little information regarding the microbial diversity and their larger role in the ecosystem functioning.

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WATER COLUMN



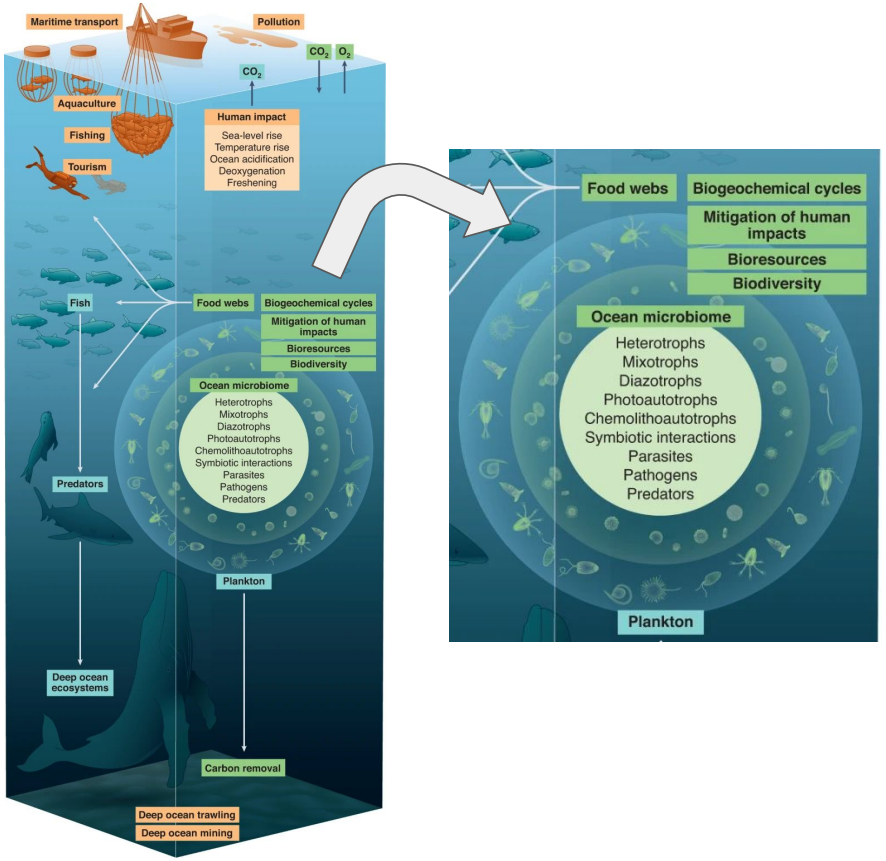
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Marine Ecosystems: Water Column

The ocean microbiome is composed of a vast assortment of prokaryotes, eukaryotic microbes and viruses, which have a range of different lifestyles driving microbial interactions.

They provide essential **ecosystem services**, such as **biogeochemical cycling**, **bioresources**, **biodiversity**, **resources** for the entire ocean food web and **mitigation** of human activities.

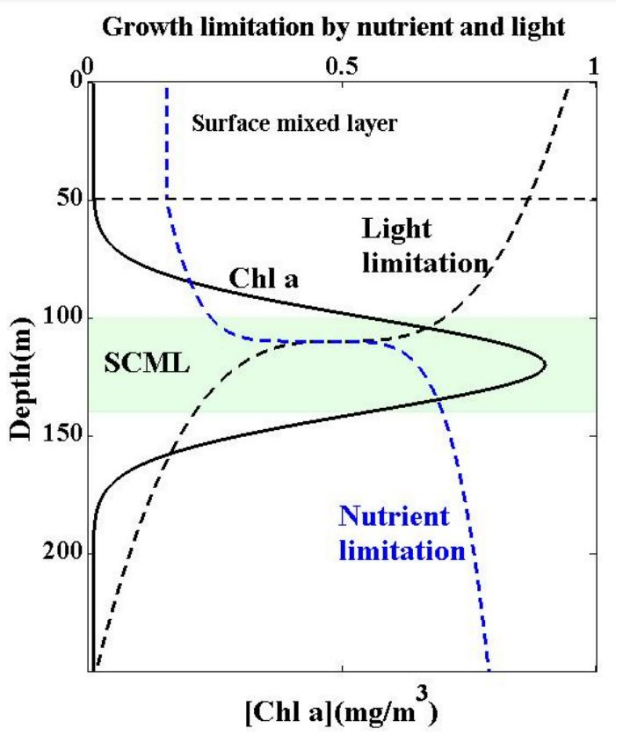
Unfortunately, human-driven processes such as increasing greenhouse gas emissions and related warming, freshening, deoxygenation and acidification of the ocean represent a huge threat to microbial communities.

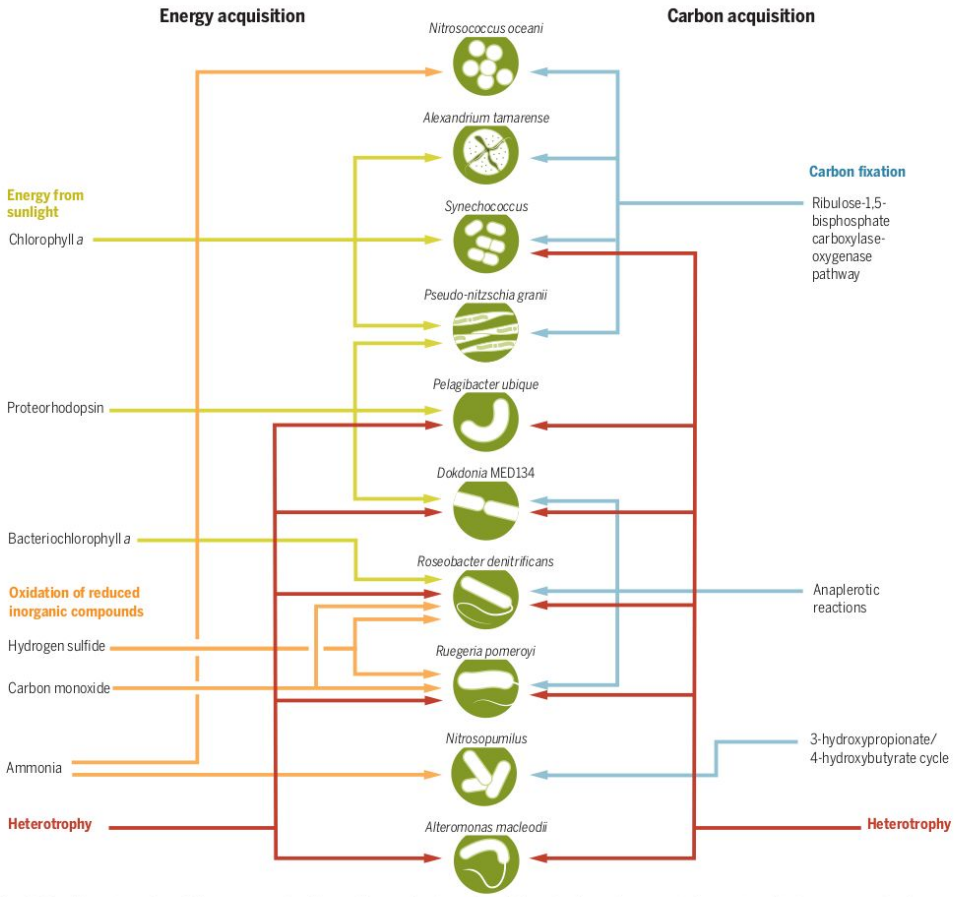


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10.1038/s41564-022-01145-5

Marine Ecosystems: Water Column





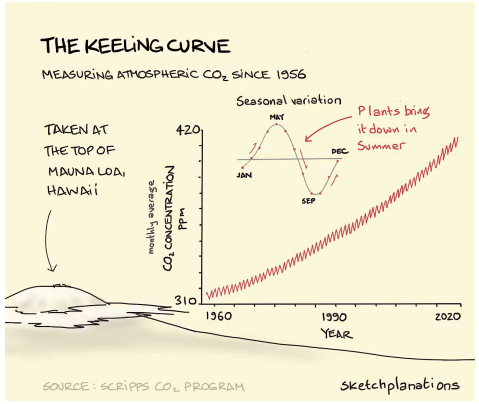
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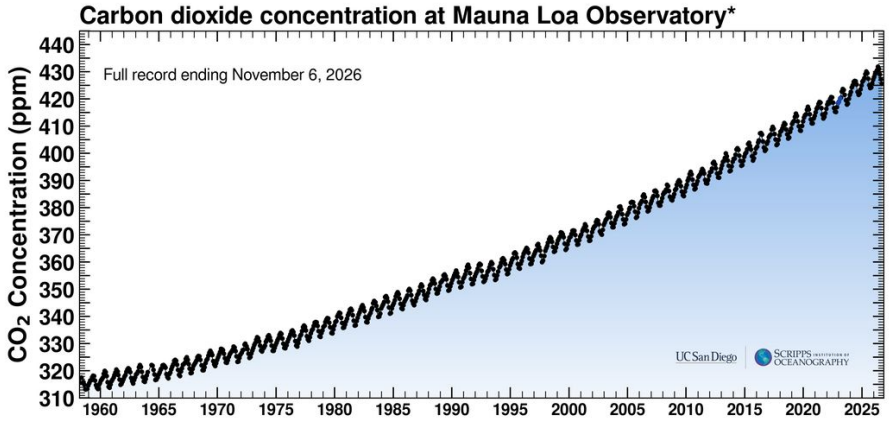
10.1126/science.aac8455

Tracking Earth's Carbon Breath

The Keeling Curve is the record of atmospheric CO_2 from the Mauna Loa Observatory, starting in 1958.





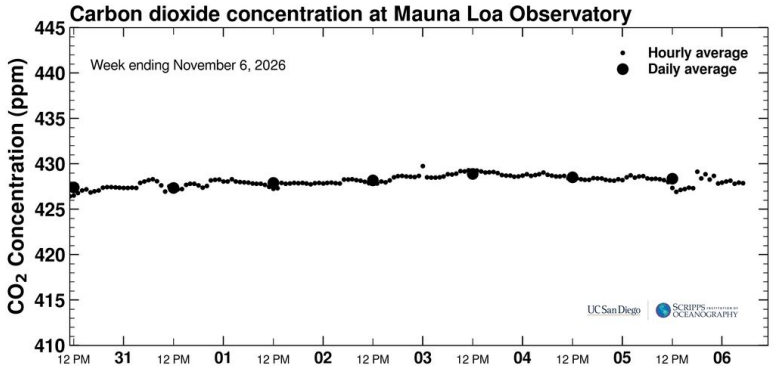


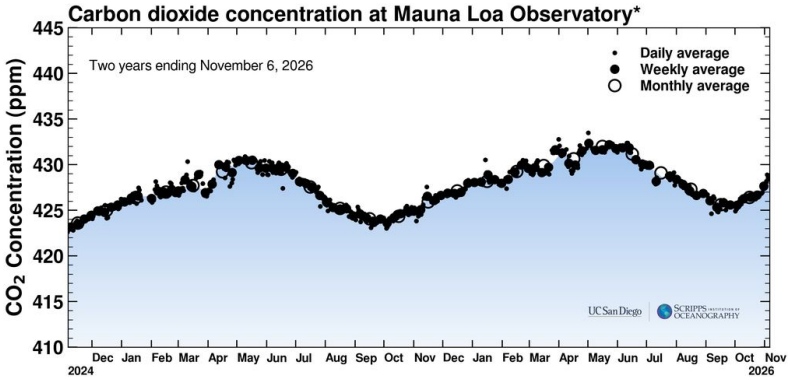
Charles Keeling's measurements show that the monthly average CO_2 concentration in the air has risen from around 310 ppm (parts per million) to around 410 ppm in 2020. As a greenhouse gas this change in CO_2 concentration is one of the critical causes of climate change.

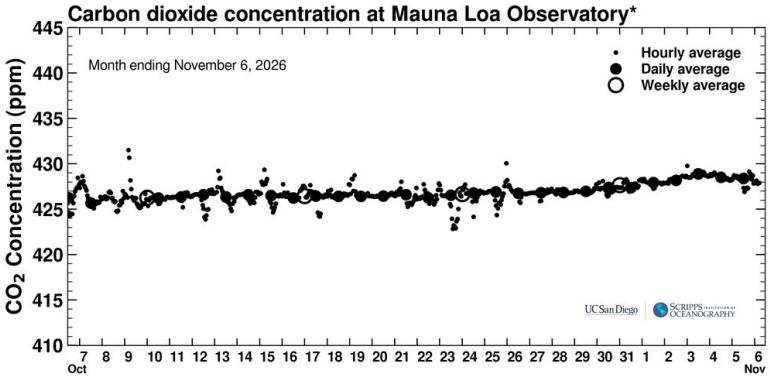
The concentration varies seasonally as plants bloom in Spring and Summer reducing the concentration from its high around May as the CO_2 has built up over the Northern Hemisphere Winter.

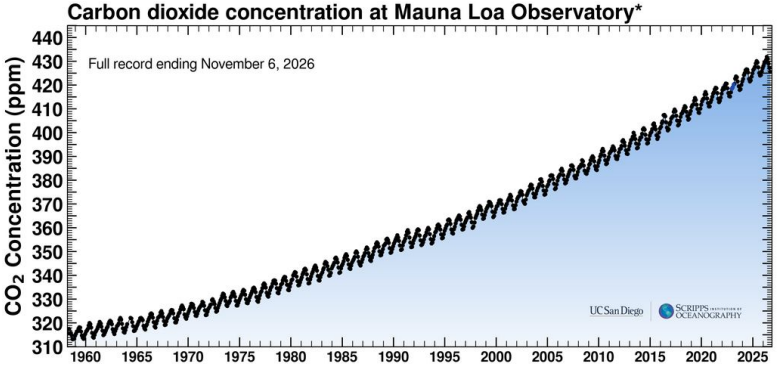
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Tracking Earth's Carbon Breath



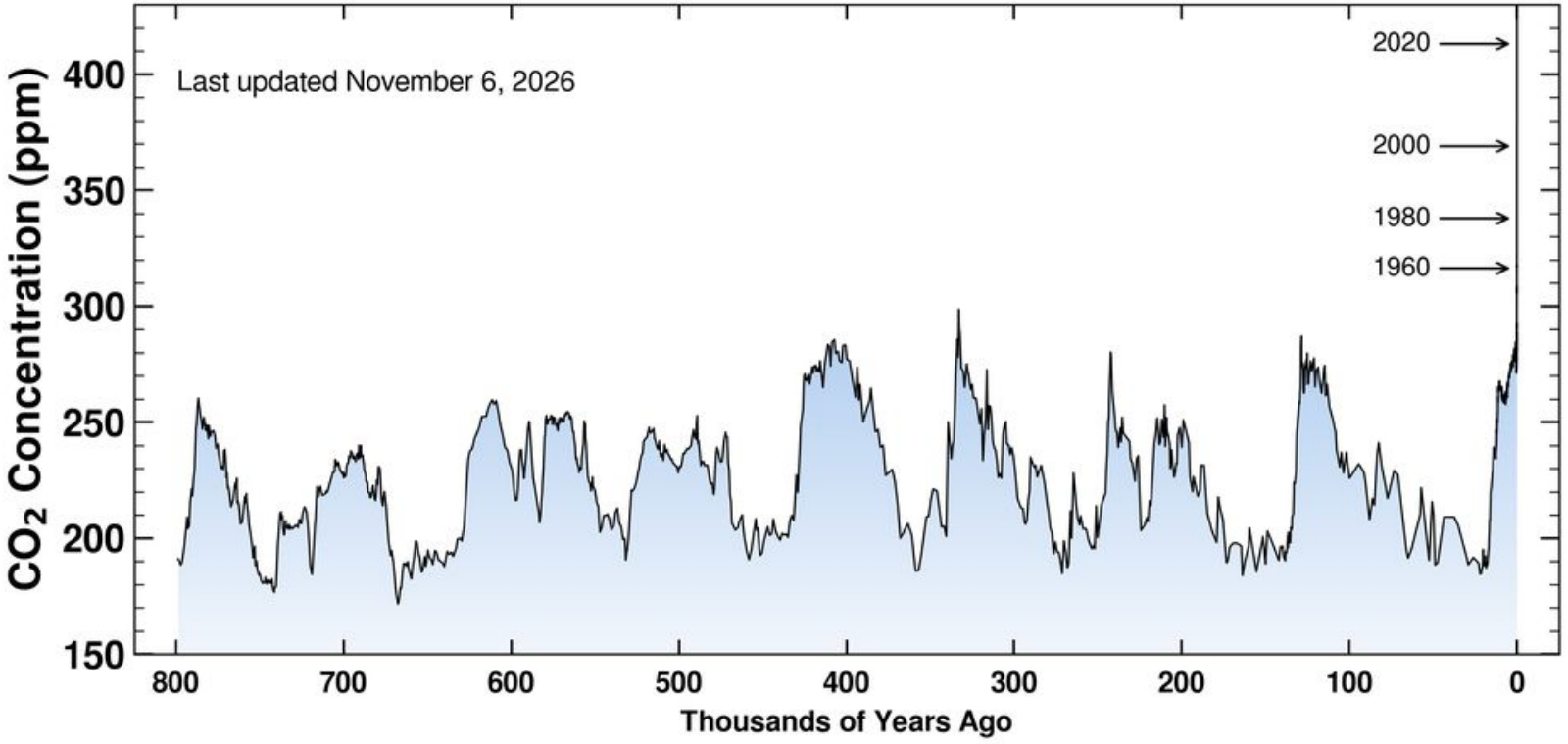






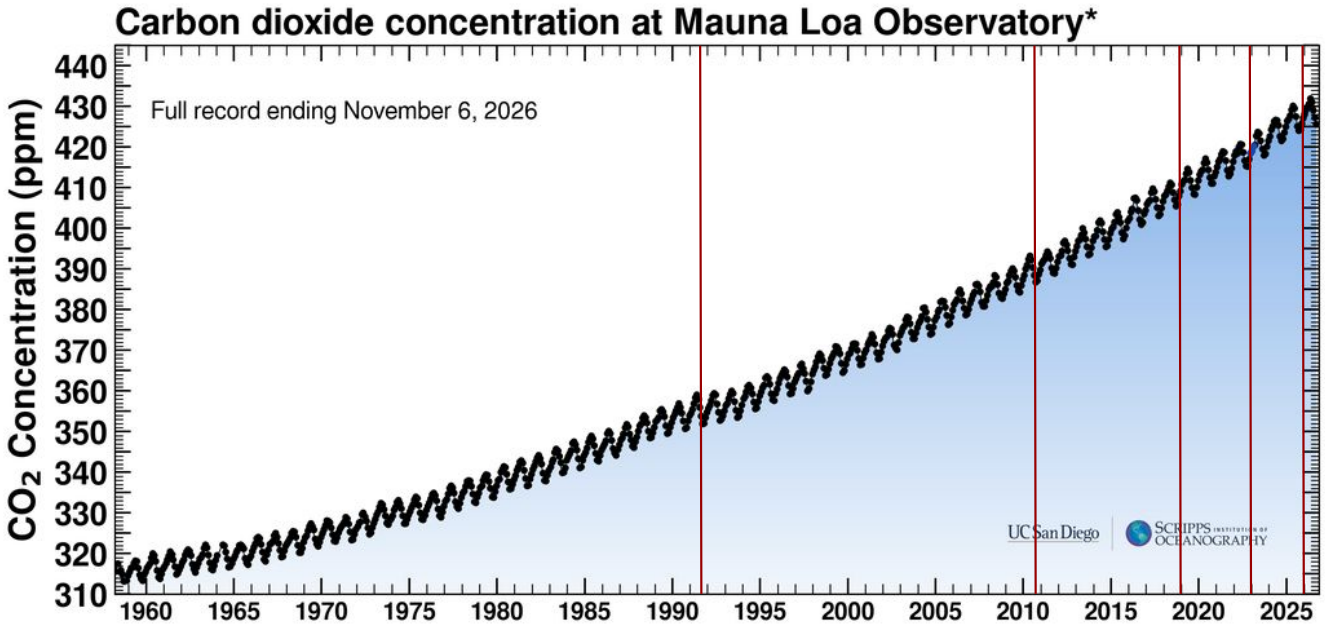
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Tracking Earth's Carbon Breath



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Keeling Curve

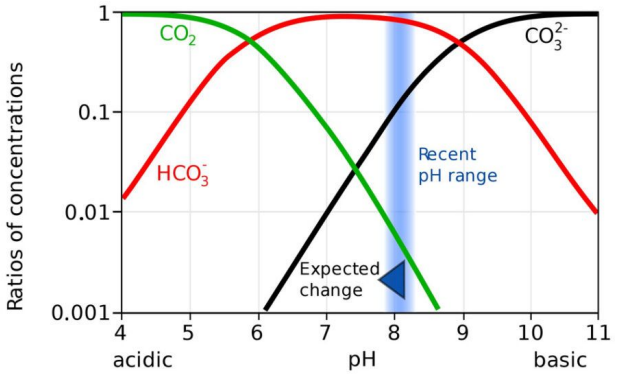


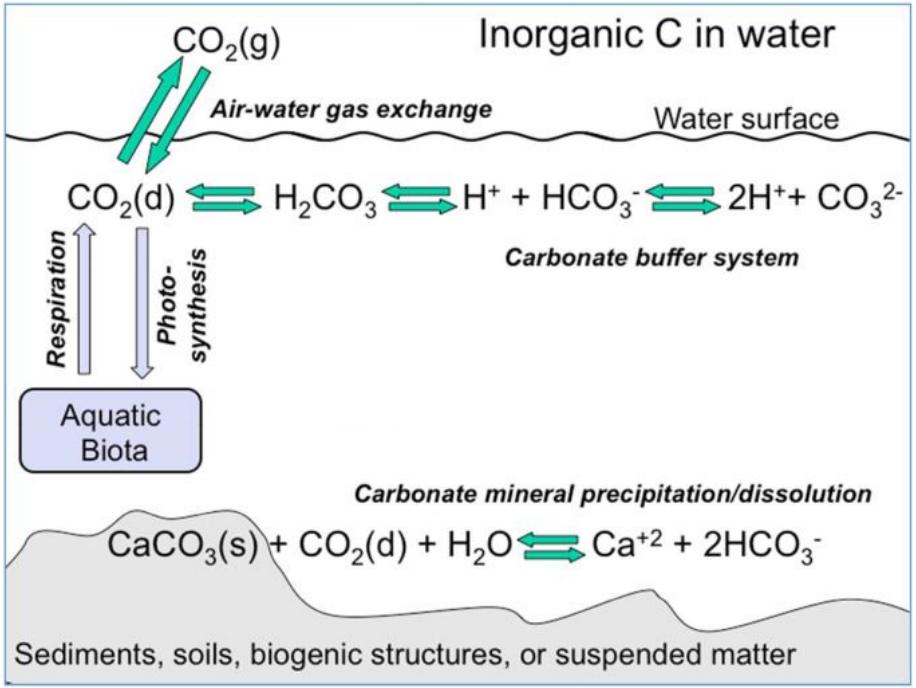
1992: 353.34 PPM
2011: 388.76 PPM
2019: 412.48 PPM
2023: 422.62 PPM
2026: 427.77 PPM

AN OVERALL AVERAGE INCREASE OF 2.2 PPM/YEAR → 17.13 $GtCO_2$/YEAR

Marine Ecosystems: Inorganic Carbon Cycle

Dissolved inorganic carbon (**DIC**) is a biogeochemically and ecologically important variable in seawaters that buffers pH, sustains photosynthesis and biogenic calcification. When CO_2 dissolves in seawater, it forms carbonic acid (H_2CO_3), which dissociates into bicarbonate (HCO_3^-) and carbonate ion (CO_3^{2-}). When atmospheric CO_2 levels rise, it promotes the formation of more carbonic acid, which releases H^+ ions, which in turn pushes the equilibrium towards bicarbonate, decreasing the availability of carbonate ions.
This latter affects organisms, like corals and mollusks, which rely on carbonate for calcium carbonate shell formation.



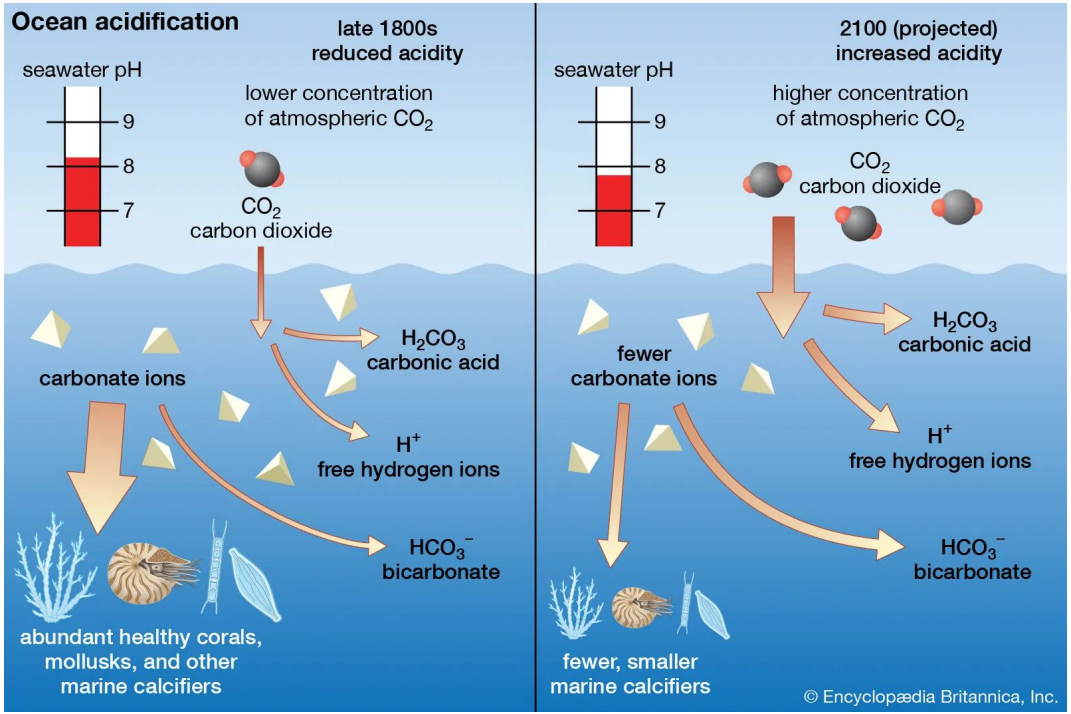


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Marine Ecosystems: Inorganic Carbon Cycle

Global-scale anthropogenic perturbations of the Earth's carbon and nutrient cycles are rapidly altering nearly every chemical, physical and biological property that affects the growth of marine microorganisms.

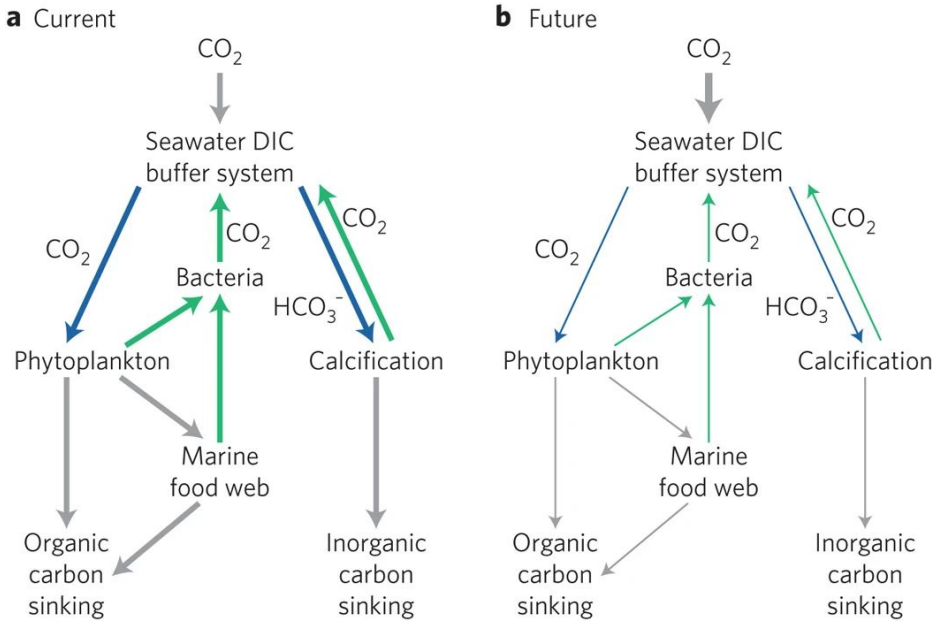
The most direct effect on the global carbon cycle is ocean acidification. Almost 1/3 of the CO_2 released to the atmosphere by fossil fuel combustion enters the surface ocean affecting the chemical equilibrium of the '**seawater carbonate buffer system**', driving down pH.



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Anthropogenic Impacts On Marine Ecosystems

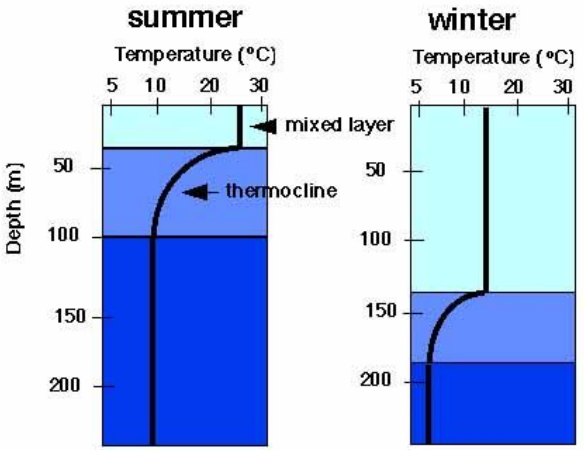
For the carbon cycle, current evidences suggest that increasing CO_2 entering the surface ocean will progressively swell the DIC, in turn lowering ocean pH. Acidification will directly inhibit calcification by most coccolithophores, thus reducing inorganic carbon sinking, and indirectly reduce carbon fixation. Consequently, there will be decreases in vertical export fluxes of organic and inorganic carbon, reduced organic carbon flow into the marine food web, lower levels of bacterial production and biomass, and smaller releases of CO_2 by microorganisms via respiration or as a byproduct of calcification.



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Focus On Ocean Stratification

Stratification in water bodies is the formation of relatively distinct and stable layers by density. It represents a **barrier** to the vertical mixing of water, which affects the exchange of heat, carbon, oxygen and nutrients. Oceans have a well-mixed surface layer where, due to wind driven movement, upwelling and downwelling phenomena can occur, in which cold nutrient-rich water rises and warm water sinks. Below the mixed layer there's the thermocline, a distinct layer with a high gradient of temperature associated with depth. Below the thermocline the deep ocean temperature is fairly constant around 2 °C, continuing down to the bottom. The density of water varies with temperature and salinity, in turn stratification varies seasonally. During the summer/spring the surface water is warmer and less dense than cold, remains at the surface of a "stable" water column because of the absence of mixing events. During the fall/winter, cooler temperatures and wind from storms cool down the surface waters, thus lowering the mixed layer.

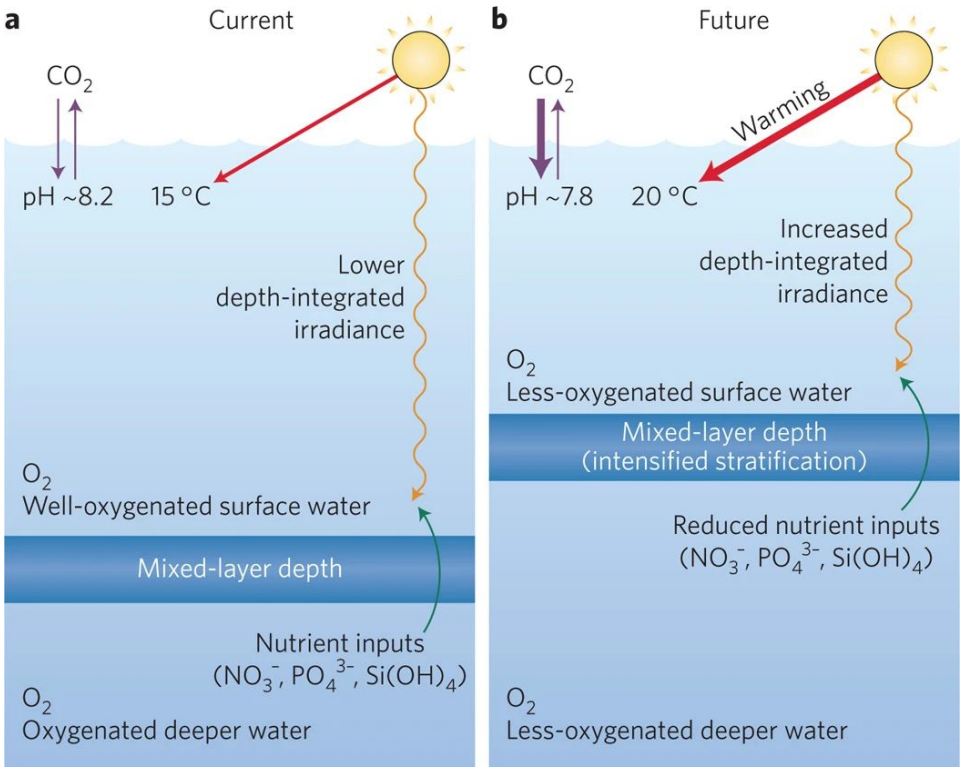


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10.1038/nmicrobiol.2017.58

Anthropogenic Impacts On Marine Ecosystems

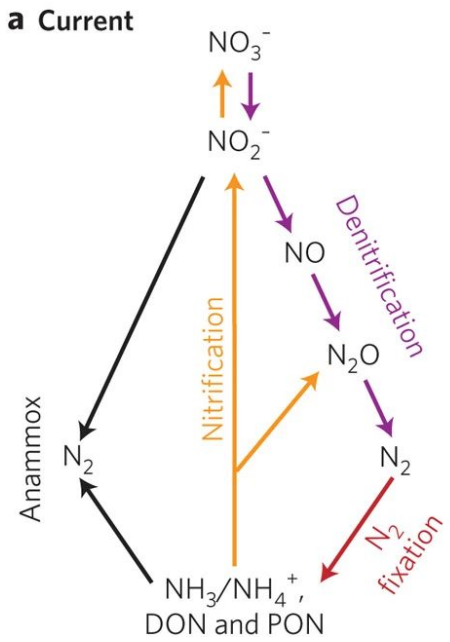
Among the changes occurring over the next 100 years that will have important impacts on marine microbiology are warming of the surface ocean due to greenhouse gases; intensified density-driven stratification, leading to lower vertical fluxes of nutrients such as nitrate, phosphate, and silicate; shallowing of the surface mixed layer, resulting in increases in depth-integrated light exposure; and reduction of seawater oxygen concentrations in both the surface mixed layer and the underlying deep ocean.

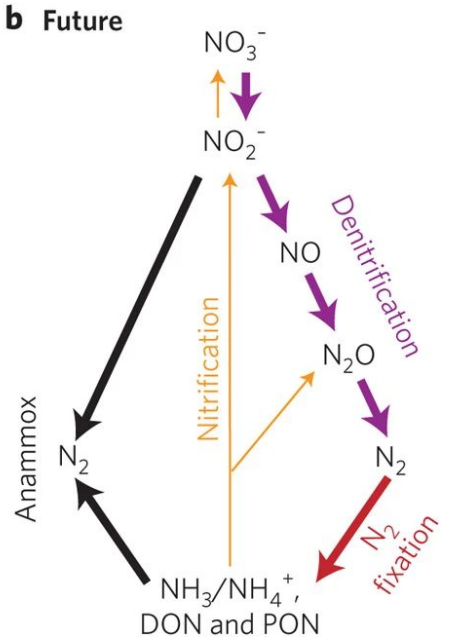


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Anthropogenic Impacts On Marine Ecosystems

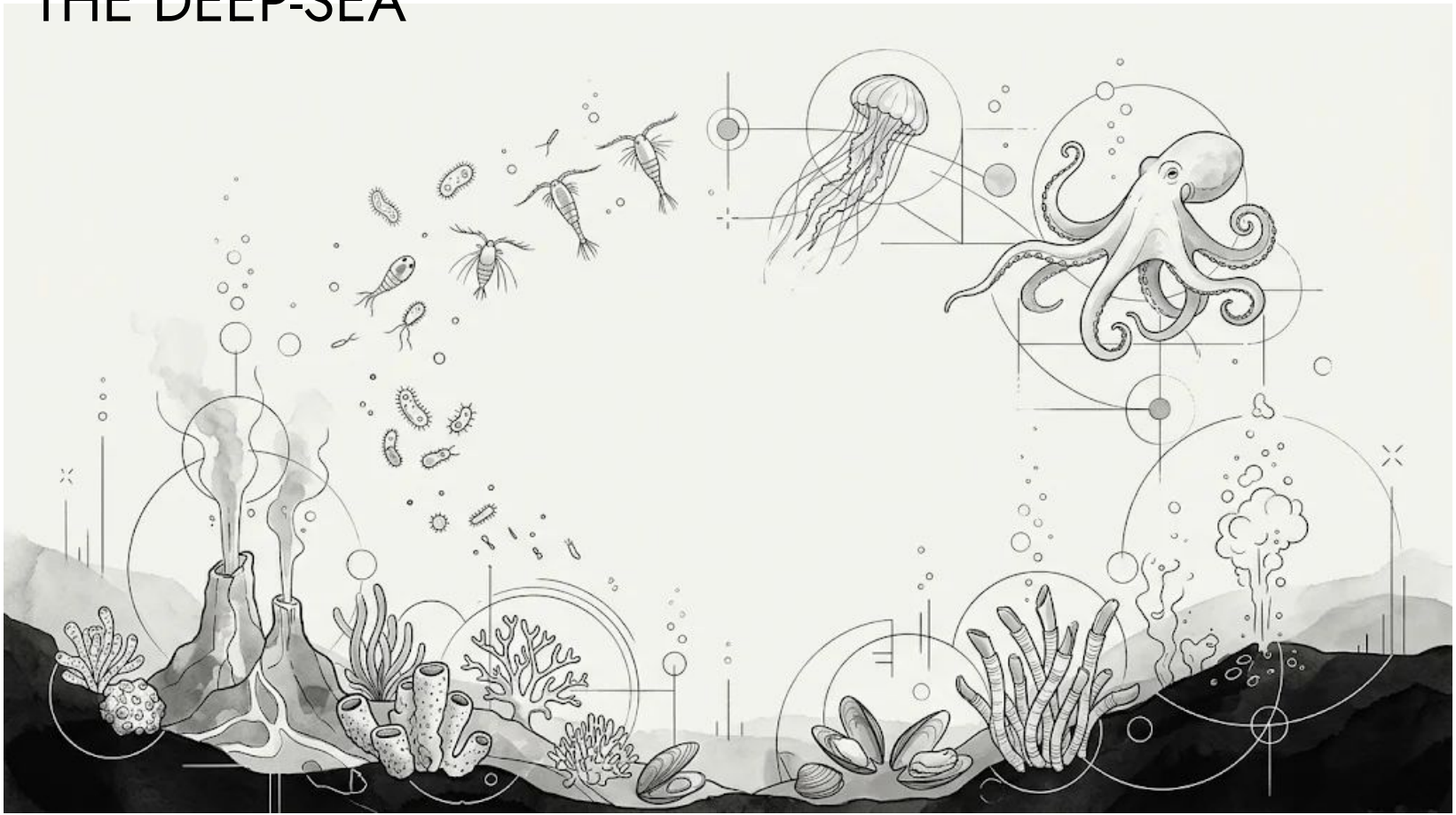
One of the most direct effects of ocean deoxygenation will be the intensification of microbial processes occurring in low-oxygen environments, such as **denitrification** and **anammox**. Nitrate will be used as an alternative to oxygen for respiration, converting it into nitrogen gas through denitrification, or combining it with ammonium in anammox reactions. The increased N_2 release into the atmosphere will effectively remove bioavailable nitrogen from the ocean system. The end result could be a limited productivity with a subsequent disruption of marine food webs.





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THE DEEP-SEA



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Marine Ecosystems: Deep-Sea

The deep sea or deep layer is the lowest layer in the ocean. Different definitions are available: deeper than 200 m, after the continental shelf break or deeper than 1000 m. Whatever definition we use, the important fact is that deep sea represents the majority of the global sea and harbors the majority of the microbial biomass in the ocean. Physical parameters change with depth; in particular, hydrostatic pressure increases linearly as depth increases (~1 bar every 10 m + 1 bar from the air column).



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Far Below the Surface of the World's Oceans, a Tough Place for Foam Cups





1500 m = 151 bar

https://www.nytimes.com/2008/03/25/science/25cups.html

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Life In The Deep-Sea

Organisms that inhabit the deep sea face three major environmental extreme conditions, **low temperature**, **high pressure**, and **low nutrient** levels. In addition, deep-sea waters are generally completely **dark** such that photosynthesis is impossible. Thus, microbes that inhabit the deep sea are generally chemotrophic and able to grow under high pressure and oligotrophic conditions in the cold.

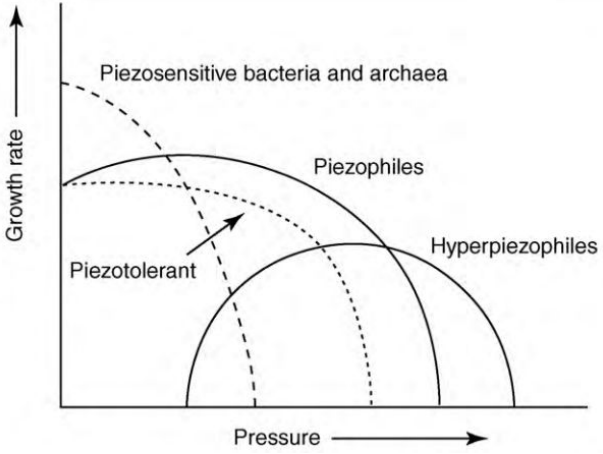


Table 1 Classification scheme of piezophiles based on optimal growth temperature and pressure[a]

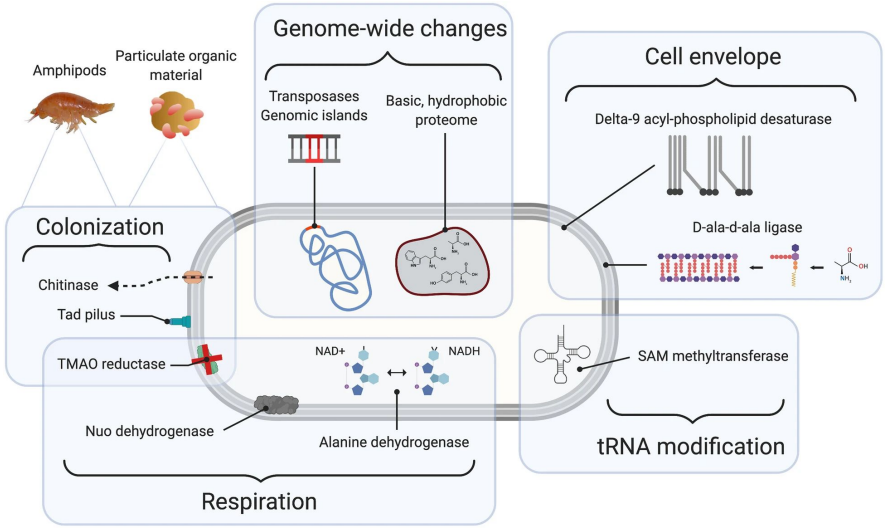
P_{kmax}|T_{kmax}	< 15 °C	15–45 °C	45–80 °C	>80 °C
Piezotolerant (< 10 MPa)	Psychro-piezotolerant	Meso-piezotolerant	Thermo-piezotolerant	Hyperthermo-piezotolerant
Piezophilic (10–50 MPa)	Psychro-piezophile	Meso-piezophile	Thermo-piezophile	Hyperthermo-piezophile
Hyperpiezophilic (> 50 MPa)	Psychro-hyperpiezophile	Meso-hyperpiezophile	Thermo-hyperpiezophile	Hyperthermo-hyperpiezophile

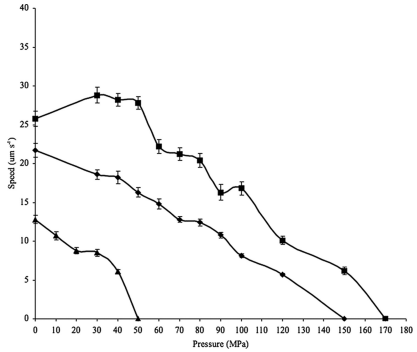
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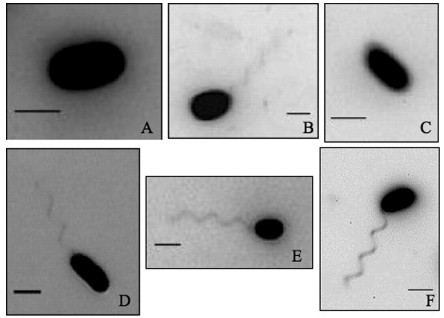
10.1016/j.tim.2010.06.006

Life In The Deep-Sea

Deep-sea microbes evolved different strategies to thrive in this habitat. For instance, in deep sea microbes developed **lateral flagella for swarming**, proteins show a higher proportion of **hydrophobic and less voluminous amino acids to stabilize the folding**, **cell membranes** have more **unsaturated fatty acids** that preserve their functionality in high pressure and low temperature conditions, and **genomes are highly plastic** to enable survival in the lack of energy availability.







Photobacterium profundum SS9 adapts to high-pressure deep-sea environments by using separate flagellar systems for swimming and swarming, maintaining an optimal growth profile at 15 °C and **28 MPa**, and structurally altering its **cell envelope**, **outer membrane proteins**, and **lipid composition** to sustain membrane fluidity.

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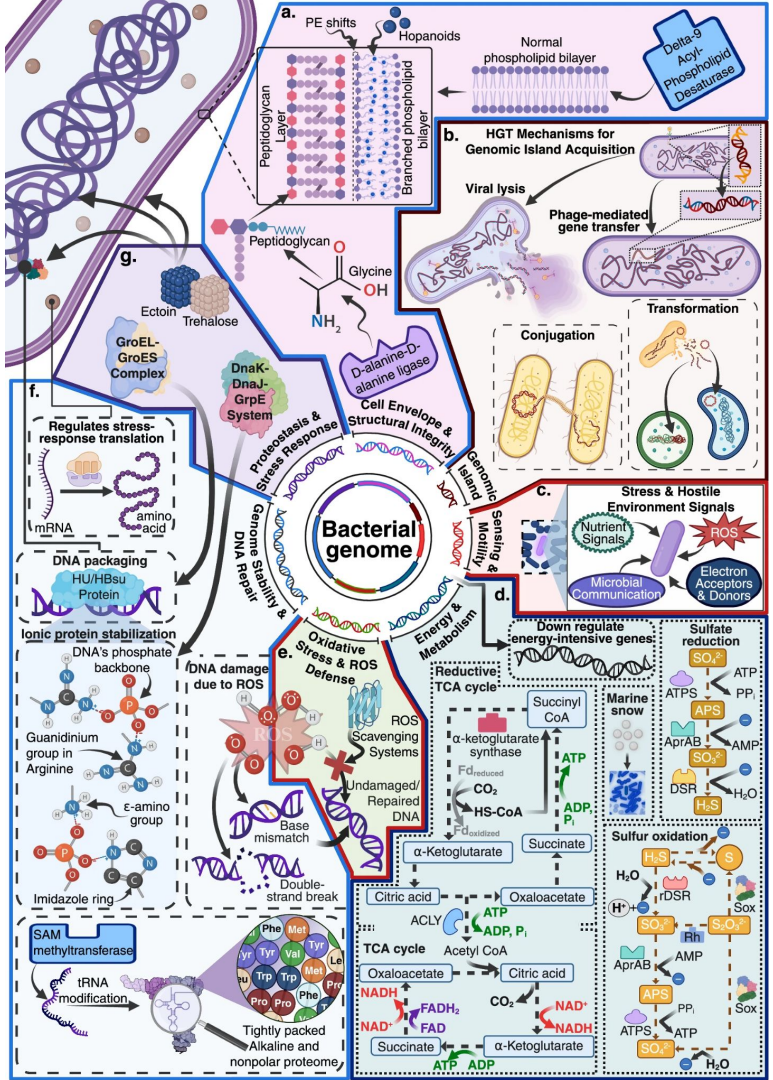
10.1186/s12864-020-07102-y

Life In The Deep-Sea

Gram-negative hadal bacteria, like Proteobacteria, survive extreme deep-sea trenches through targeted genetic and structural adaptations.

They **maintain membrane fluidity** under high pressure by altering cell envelope lipids and structures, while utilizing **horizontal gene transfer** (>> mobile genetic elements and genomic islands) to acquire traits for pressure tolerance, metal resistance, and metabolic functions. To optimize resources in cold, low-nutrient environments, they employ **environmental sensing networks**, **downregulate energy-intensive genes**, and **pivot to carbon-conserving** (glyoxylate-bypass and chemolithoautotrophy) and **anaerobic metabolic pathways** (NO_3 and SO_4 reduction).

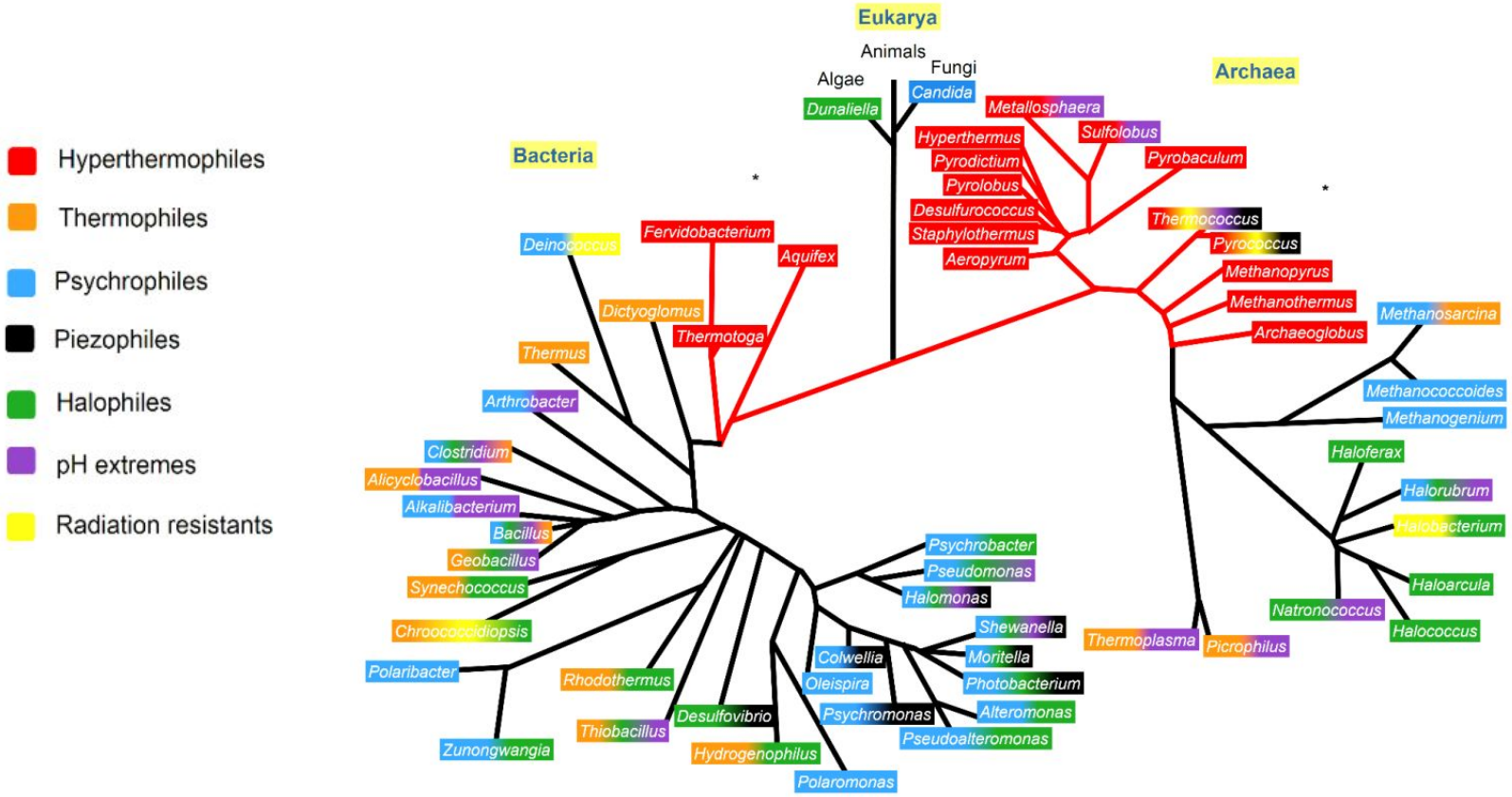
Lastly, they mitigate environmental stress and reactive oxygen species (ROS) damage using **robust antioxidant systems and DNA repair mechanisms**.



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https://doi.org/10.1038/s43247-026-03922-0

Deep-Sea Survival: When Monotolerance Isn't Enough



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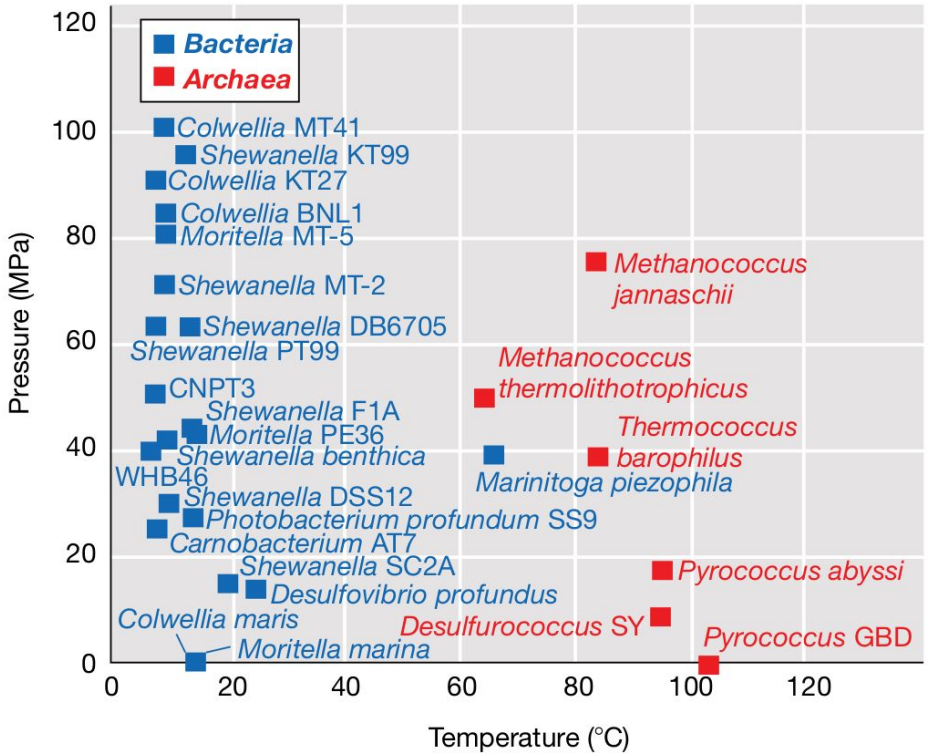
10.3390/ijms13041925

Deep-Sea Survival: When Monotolerance Isn't Enough

The oceans, which have an average depth of 4000 m and an **average pressure of 380 atmosphere (atm)** or 38 MPa, make up ~95% of the biosphere.

In the deepest parts of the oceans, pressures of 700 to 1100 atm (70 to 110 MPa) prevent the growth of most microorganisms. Moreover, the temperature in the deep sea is typically within the 1-3°C range.

This is the reason why piezophiles include species that are well adapted to high pressure and low temperatures. In addition, some of them can be also thermophilic, especially those isolated from deep-sea hydrothermal vents.



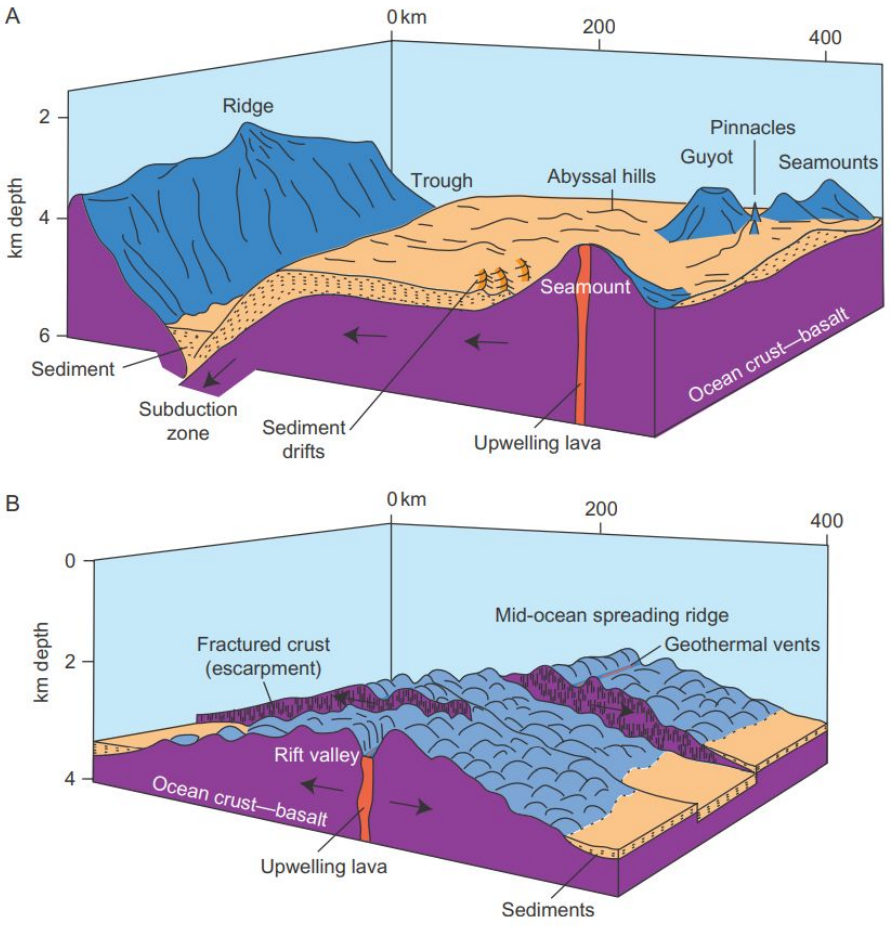
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Seafloor Geomorphology

The seafloor is not totally flat but shows peculiar **geomorphic features**, such as submarine **canyons**, **seamounts**, **spreading ridges**, **escarpments**, **plateaus**, **trenches** etc., that are commonly **associated with particular suites of habitats and biological communities**.

Benthic habitat maps can illustrate the nature, distribution, and extent of distinct physical environments and importantly, they can predict the distribution of the associated species and communities.

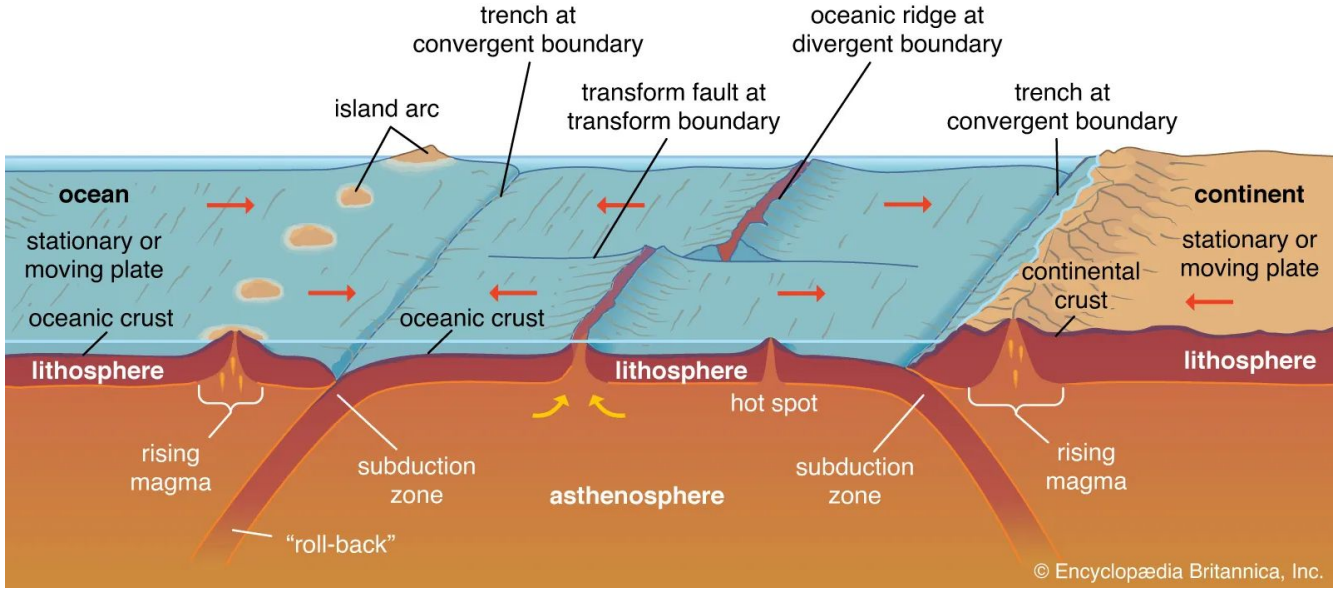
The natural diversity of geological features has been termed **geodiversity**, and the conservation of such diversity is essential in sustaining the biodiversity of an ecosystem.



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10.1016/B978-0-12-385140-6.00006-2

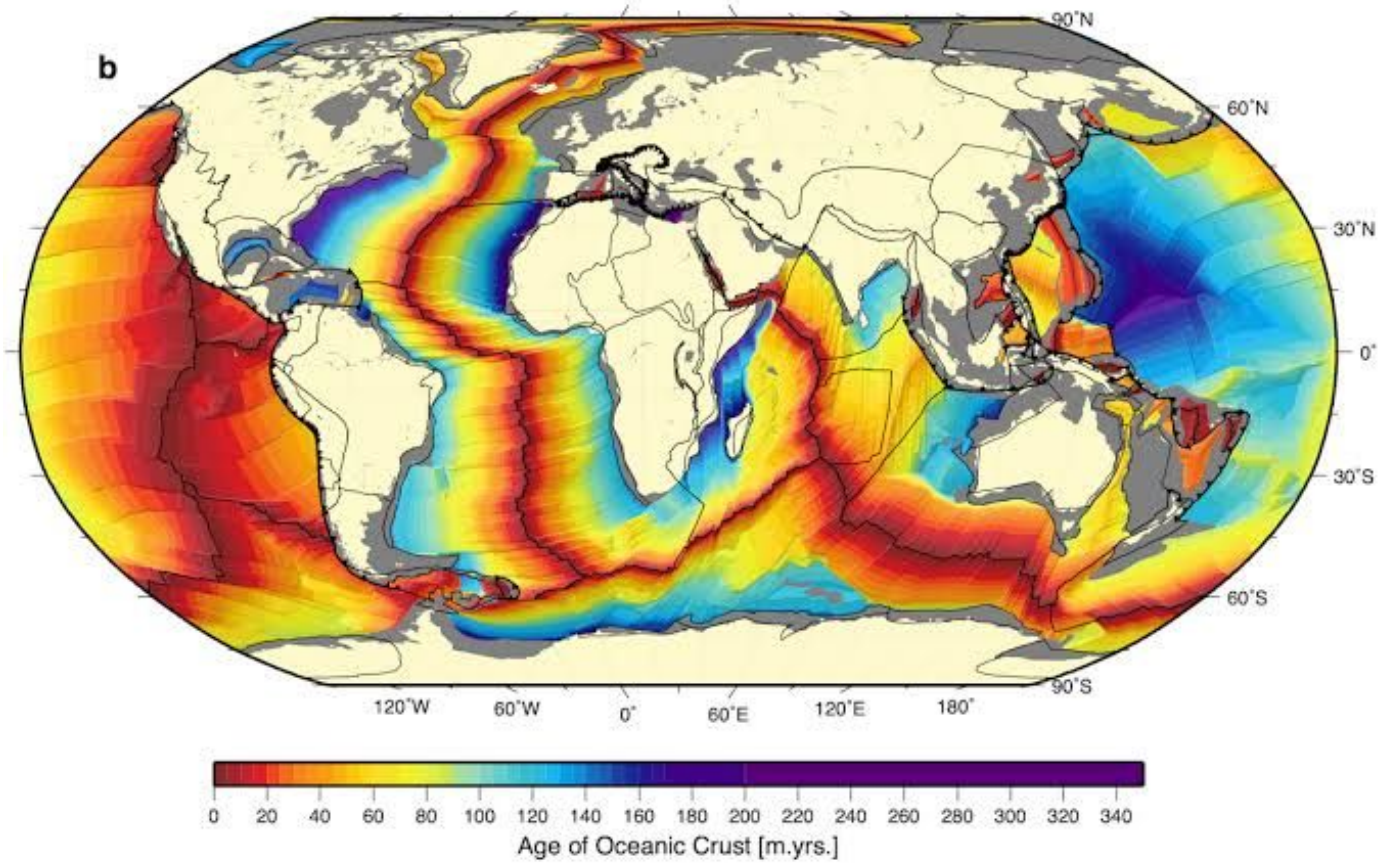
Plate Tectonic and Seafloor



The deep sea or deep layer is the lowest layer in the ocean. Different definitions are available: deeper than 200 m, after the continental shelf break or deeper than 1000 m. Whatever definition we use, the important fact is that deep sea represents the majority of the global sea and harbors the majority of the microbial biomass in the ocean. In the depth some physical parameters change, especially pressure, which increases at increasing depth.

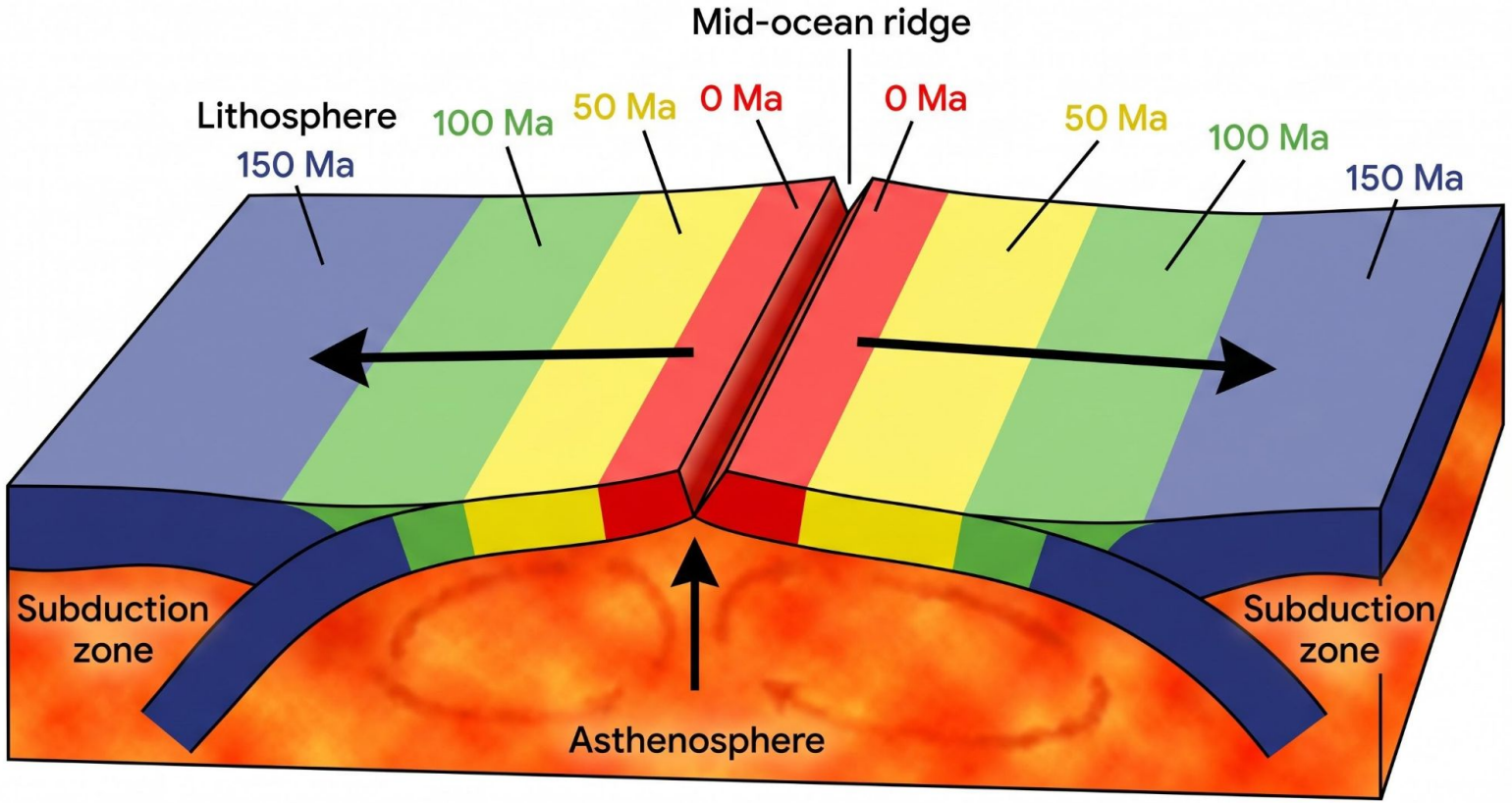
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Plate Tectonic And Seafloor



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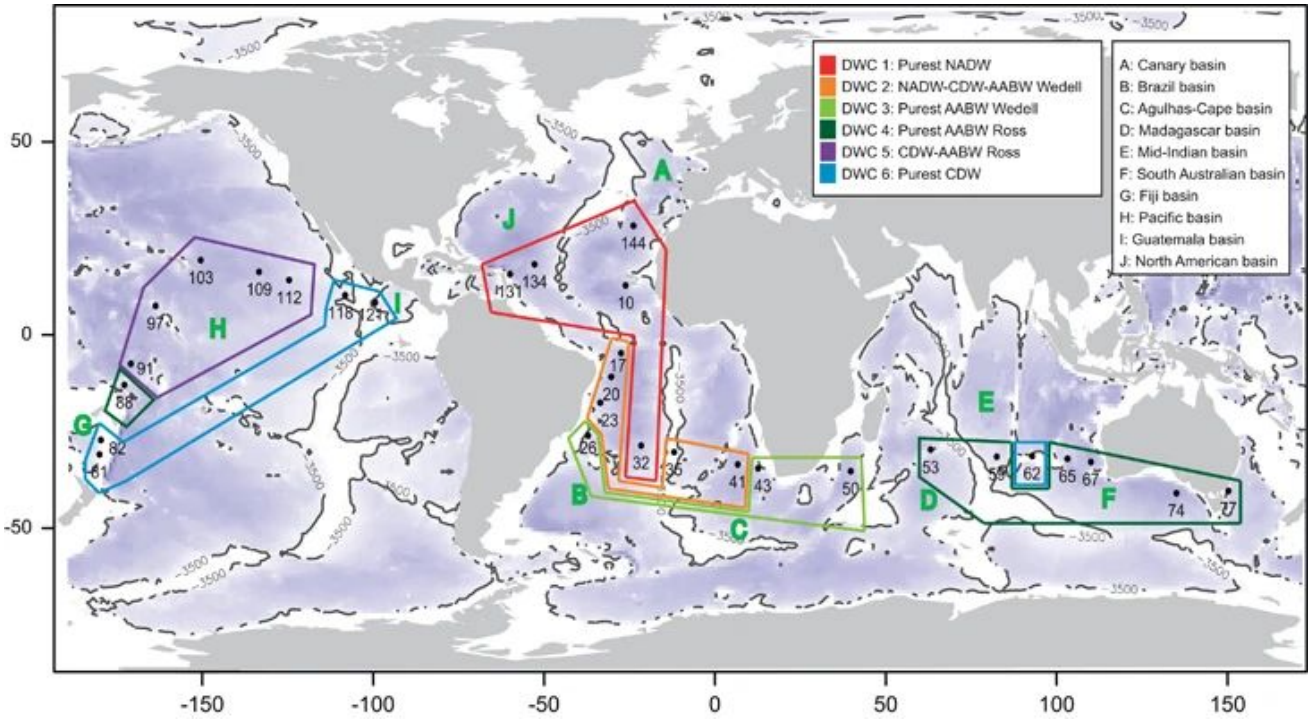
Plate Tectonic And Seafloor



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Global Diversity And Biogeography Of Deep-Sea Pelagic Prokaryotes

A globally comprehensive survey of the prokaryotic communities inhabiting the bathypelagic ocean (1000-4000 m depth) was carried out during the **Malaspina 2010 circumnavigation expedition** to describe their diversity, community structure and biogeographical distribution.

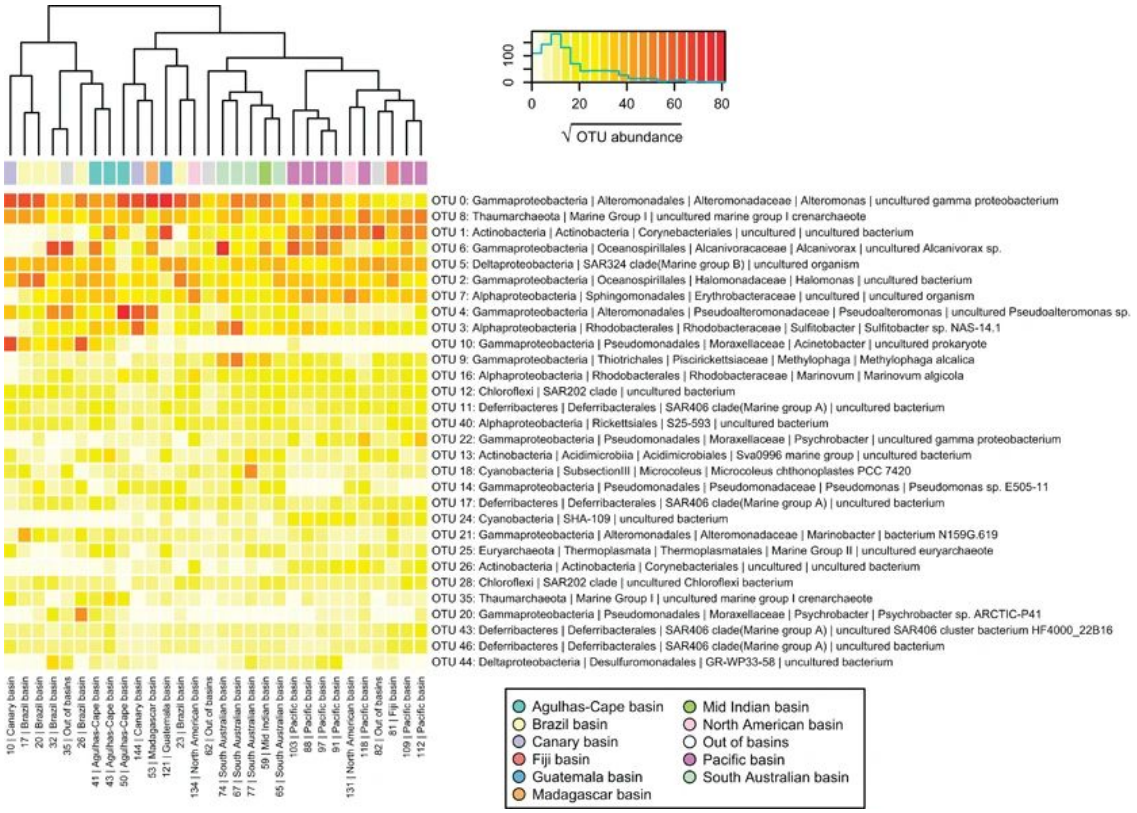


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10.1038/ismej.2015.137

Global Diversity And Biogeography Of Deep-Sea Pelagic Prokaryotes

Few specific microbial groups dominated in all the stations (relative abundance), often found in low-oxygen, high-pressure conditions. **Gamma-, Alpha- and Delta-proteobacteria** involved in organic matter degradation, sulfur cycling, and nitrate reduction. **Actinobacteria**, specialized in degrading complex organic compounds, contributing to carbon cycling. **Thaumarchaeota**, primarily ammonia-oxidizers, key players in nitrogen cycling.

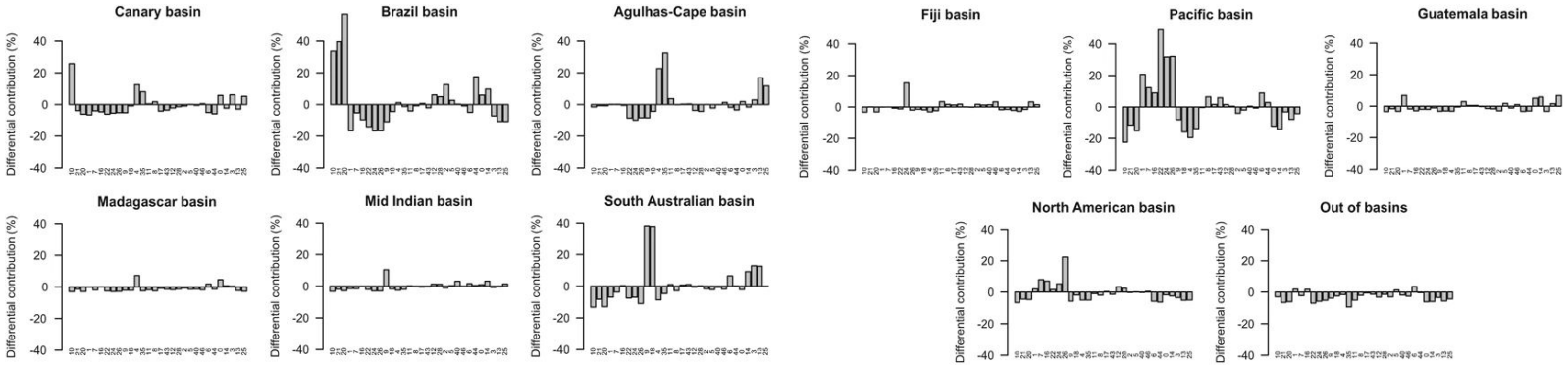


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10.1038/ismej.2015.137

Global Diversity And Biogeography Of Deep-Sea Pelagic Prokaryotes

Particle-attached communities (**seston epipelon**) appear to respond to a complex set of processes where the ageing and global circulation of the water masses and some degree of dispersal limitation create **basin-specific communities**. A fraction of the deep-oceanic particles where the prokaryotes are associated, instead of coming from the surface ocean through sinking, correspond to presumably buoyant or slow-sinking particles that are produced autochthonously at depth. The combination of the aging of the water masses and a lack of complete dispersal of microbial species are identified as the main drivers for this biogeographical pattern.

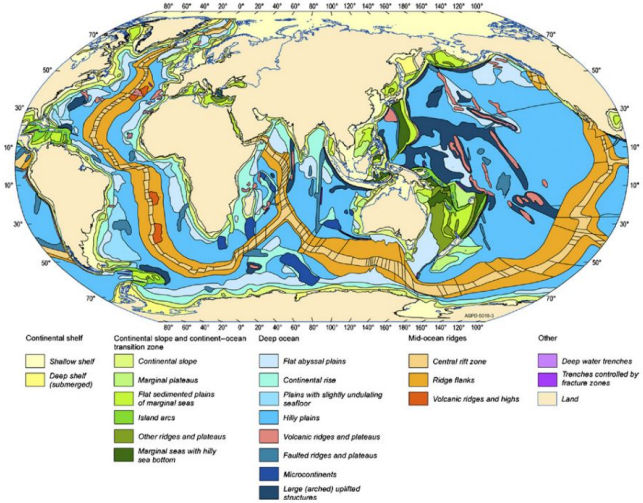


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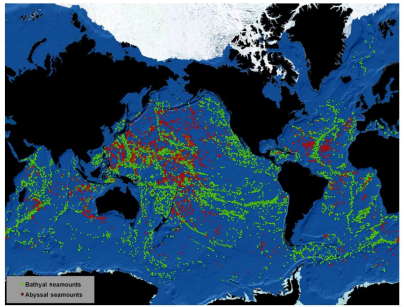
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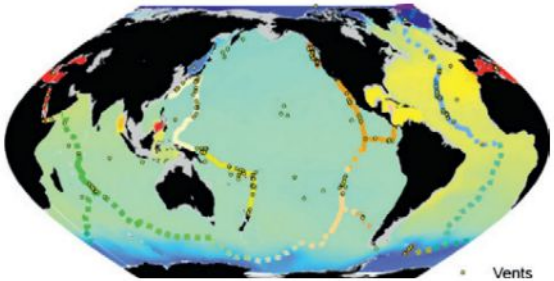
Seafloor Geomorphology

Constructing habitat maps provide fundamental information that can be used for a range of **management of ecosystem services and industrial applications**, including the management of fisheries, spatial marine environmental management, design of marine reserves, supporting offshore oil and gas infrastructure development, port and shipping channel construction, maintenance dredging, tourism, and seabed aggregate mining.





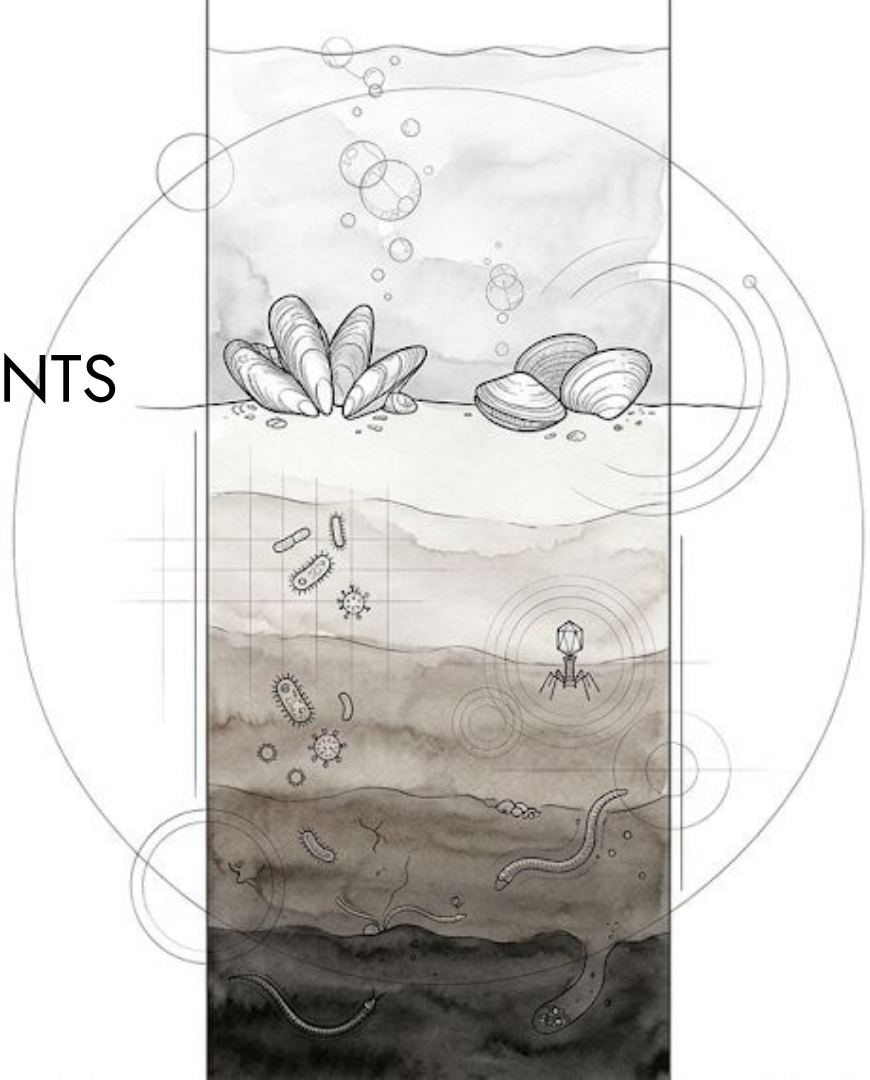




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10.1016/B978-0-12-385140-6.00006-2

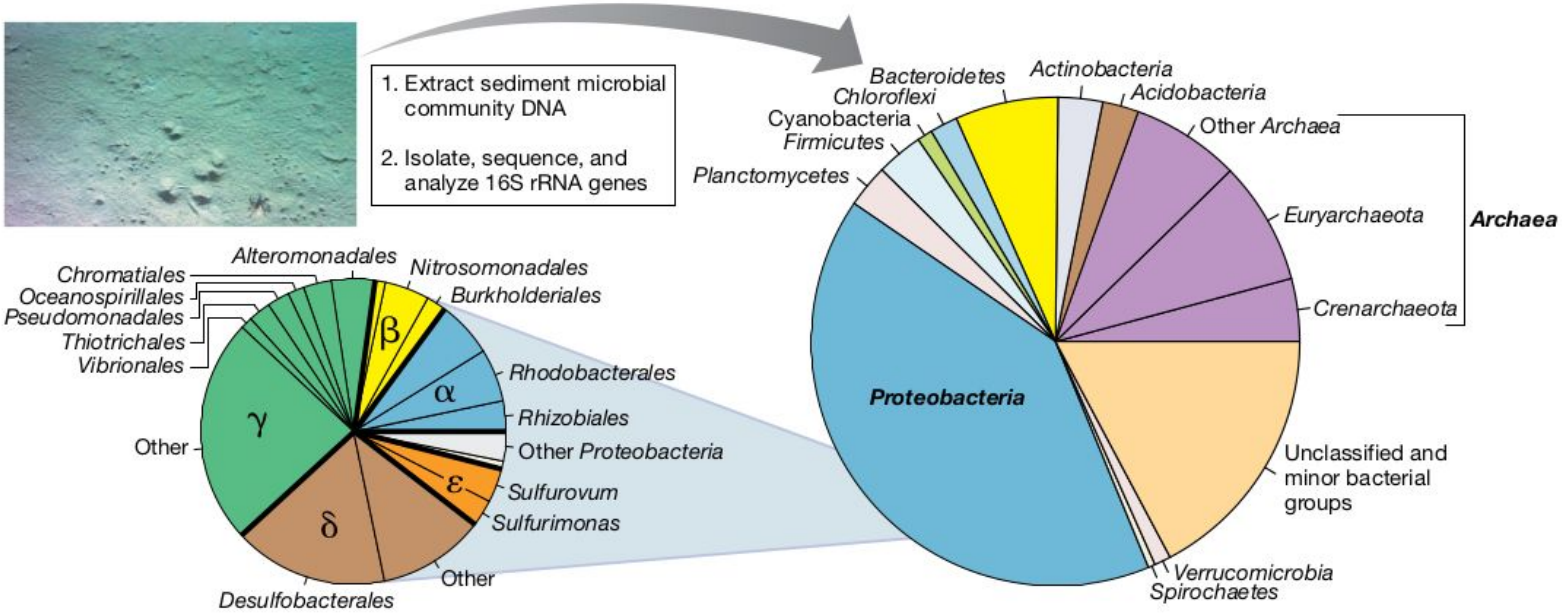
MARINE SEDIMENTS



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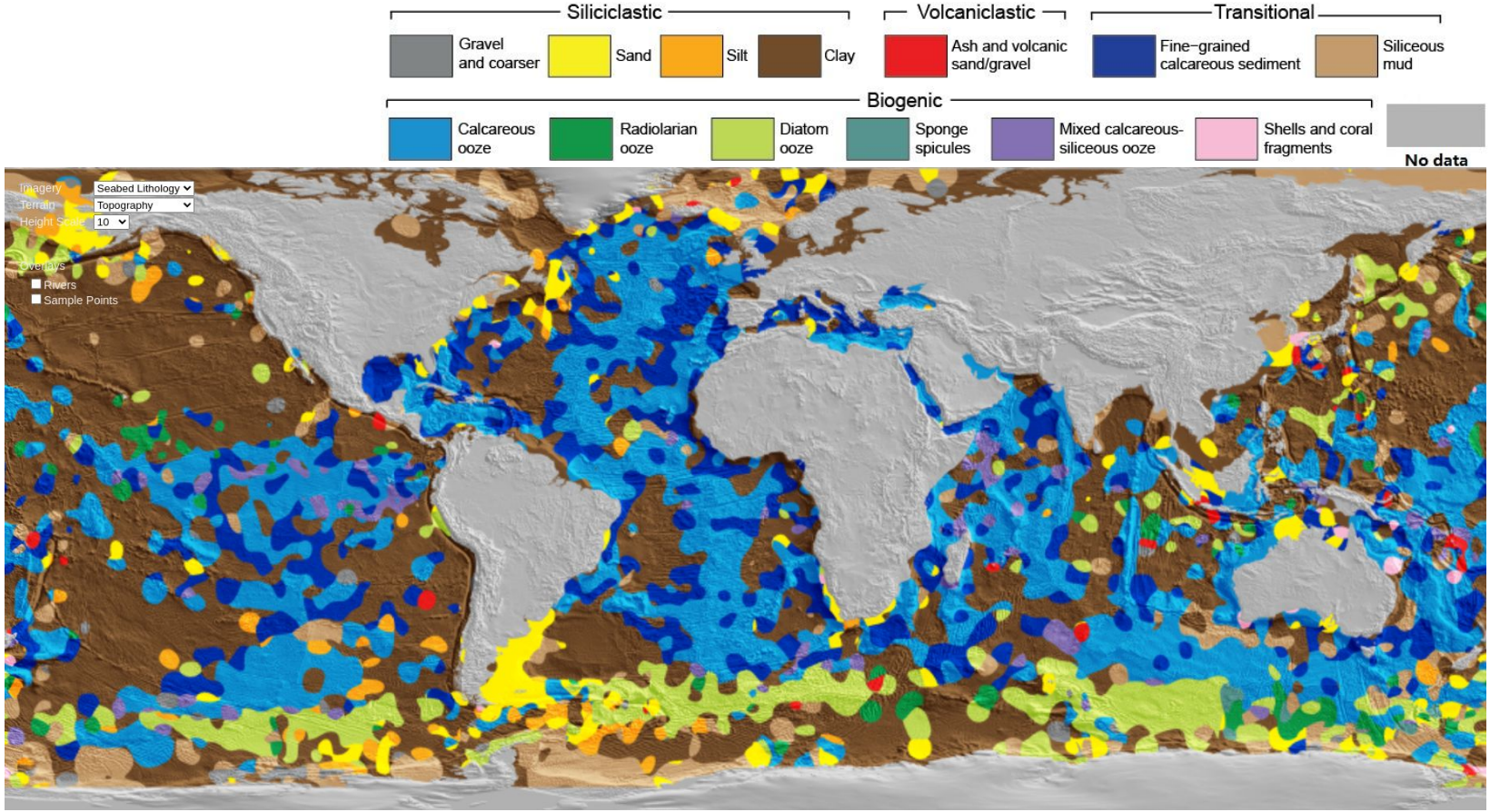
Marine Ecosystems: Surface And Deep-Sea Sediments

Another vast and mostly unexplored microbial biosphere exists below the seafloor in the marine sediments. The total number of microbial cells in marine sediment is presently estimated as **2.9 to 5.4 × 10^{29} cells**, accounting for up to **3.6% of Earth's total living biomass**. Marine sediment can range in thickness from a few millimetres to several tens of kilometres.



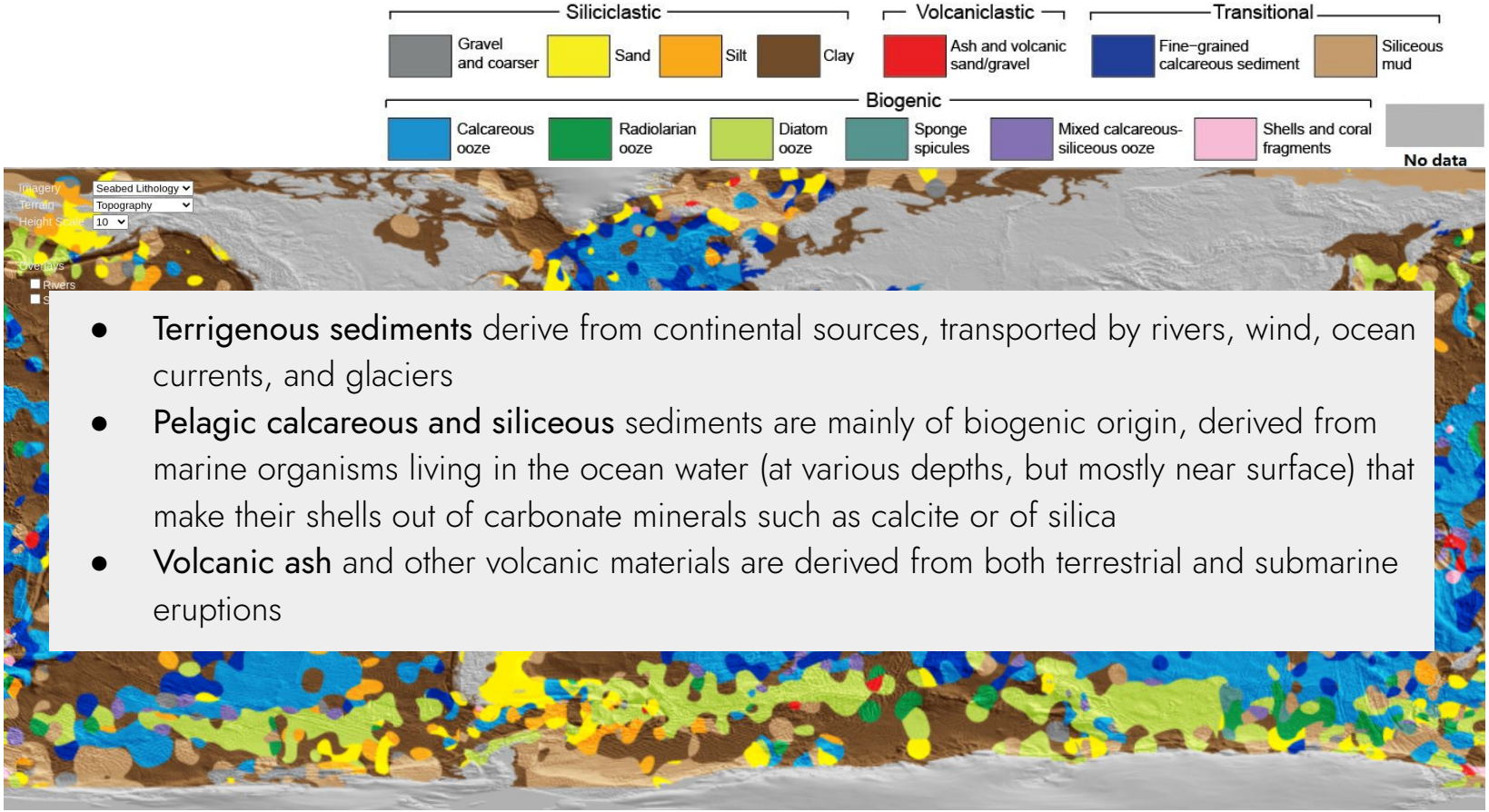
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Sediments Lithology



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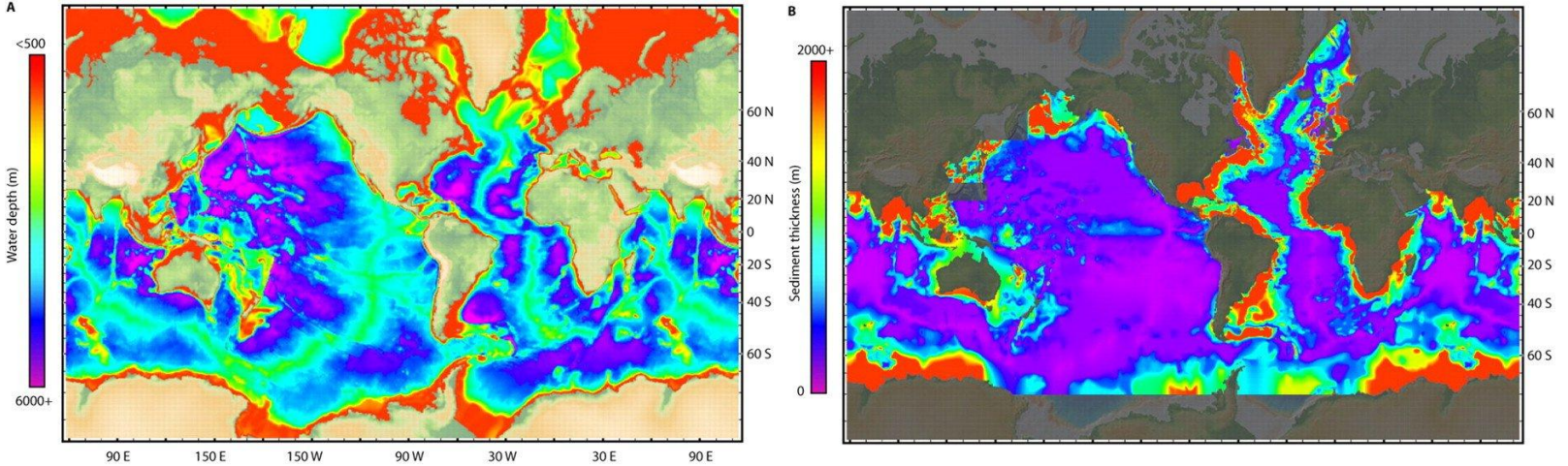
Sediments Lithology



- **Terrigenous sediments** derive from continental sources, transported by rivers, wind, ocean currents, and glaciers
- **Pelagic calcareous and siliceous** sediments are mainly of biogenic origin, derived from marine organisms living in the ocean water (at various depths, but mostly near surface) that make their shells out of carbonate minerals such as calcite or of silica
- **Volcanic ash** and other volcanic materials are derived from both terrestrial and submarine eruptions

Marine Ecosystems: Surface And Deep-Sea Sediments

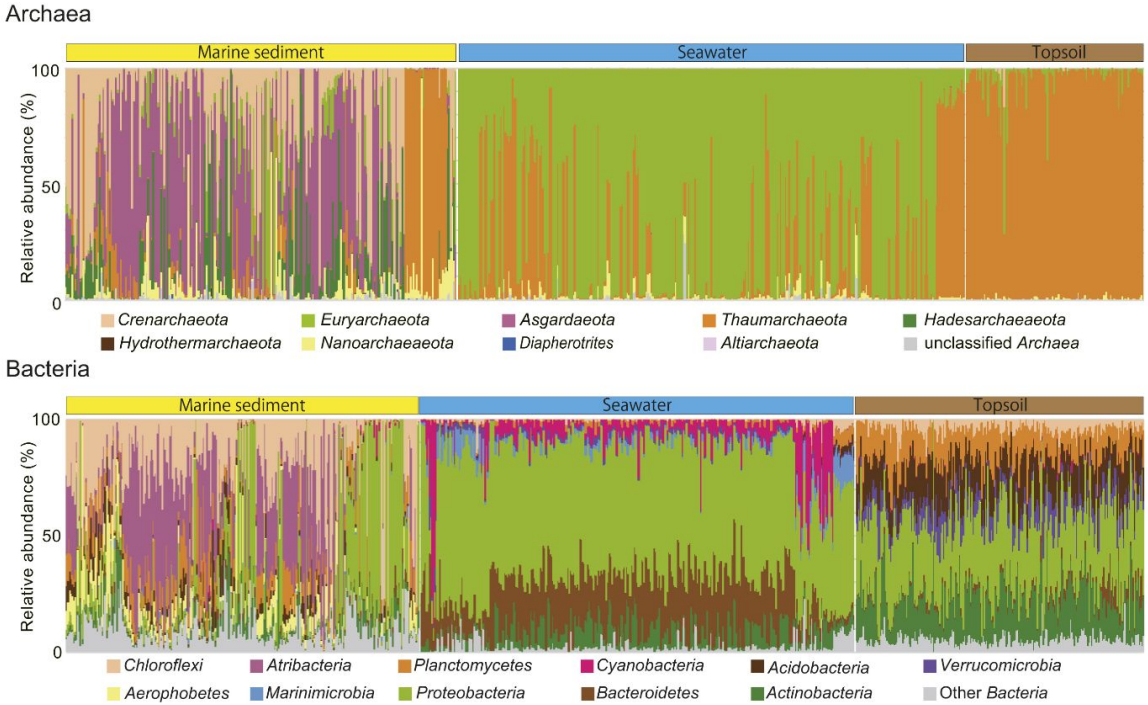
The **thickness of the sediments is** not **linked** to the water depth but rather **to the nearness to the continent**. This is because the most sediments derived from continents, so they accumulated mostly in the zones close to the coasts. A higher thickness of the sediment can be observed in the east coast of Americas respect to the west coast, due to the rivers, that flow more toward the east coast, while the west side of Americas is a **convergent margin of a subduction zone**, and therefore a mountainous area.



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Marine Sediments Microbiome

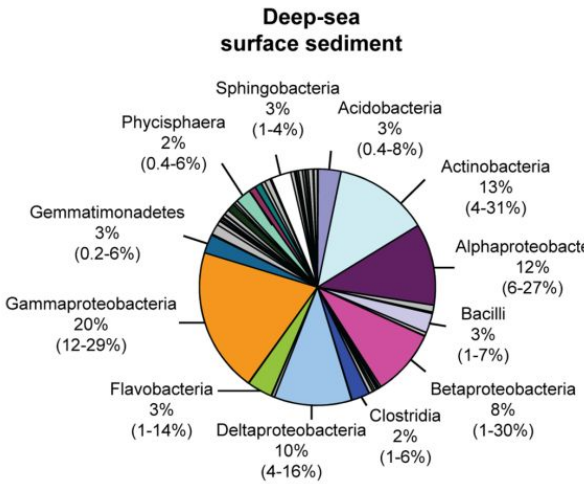
The taxonomic diversity of the marine sedimentary microbial biome and the spatial distribution of that diversity is significantly correlated with **sedimentary organic carbon concentration**, and **presence** or **absence** of **dissolved oxygen**.



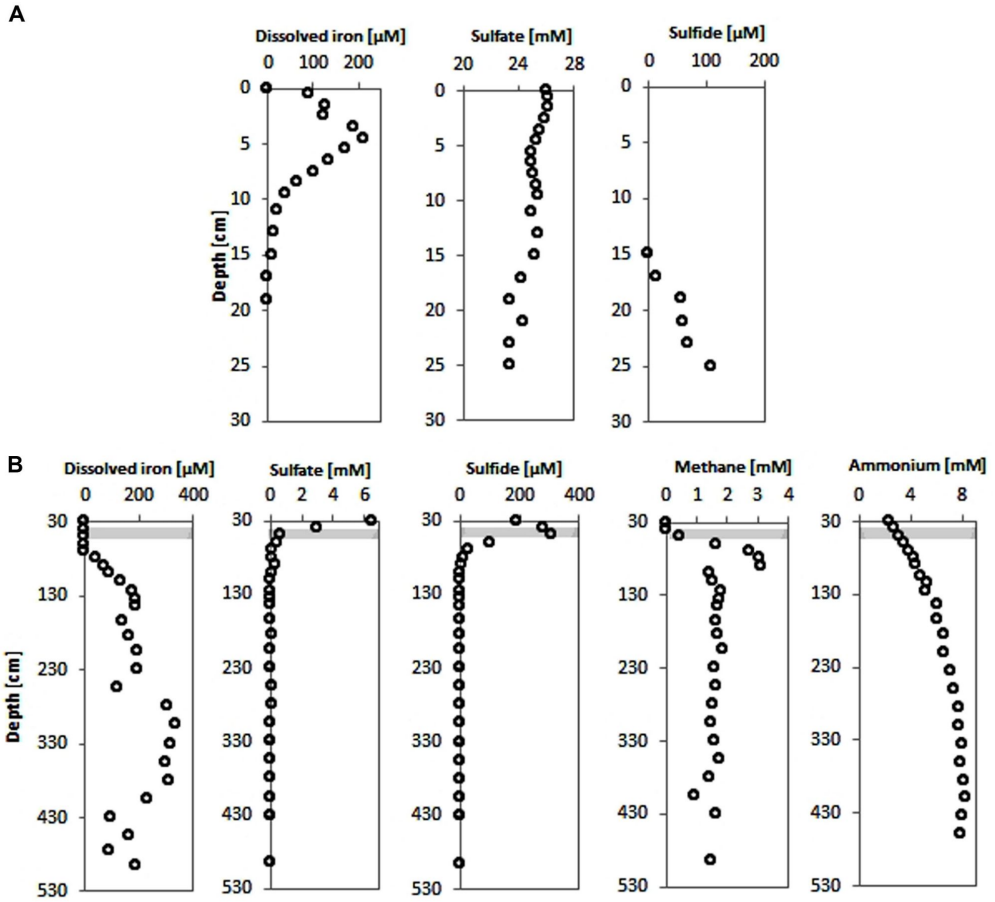
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10.1073/pnas.1919139117

Marine Sediments Microbiome



Microbial diversity and distribution in marine sediments also depends on the **availability of electron donor and acceptors**.

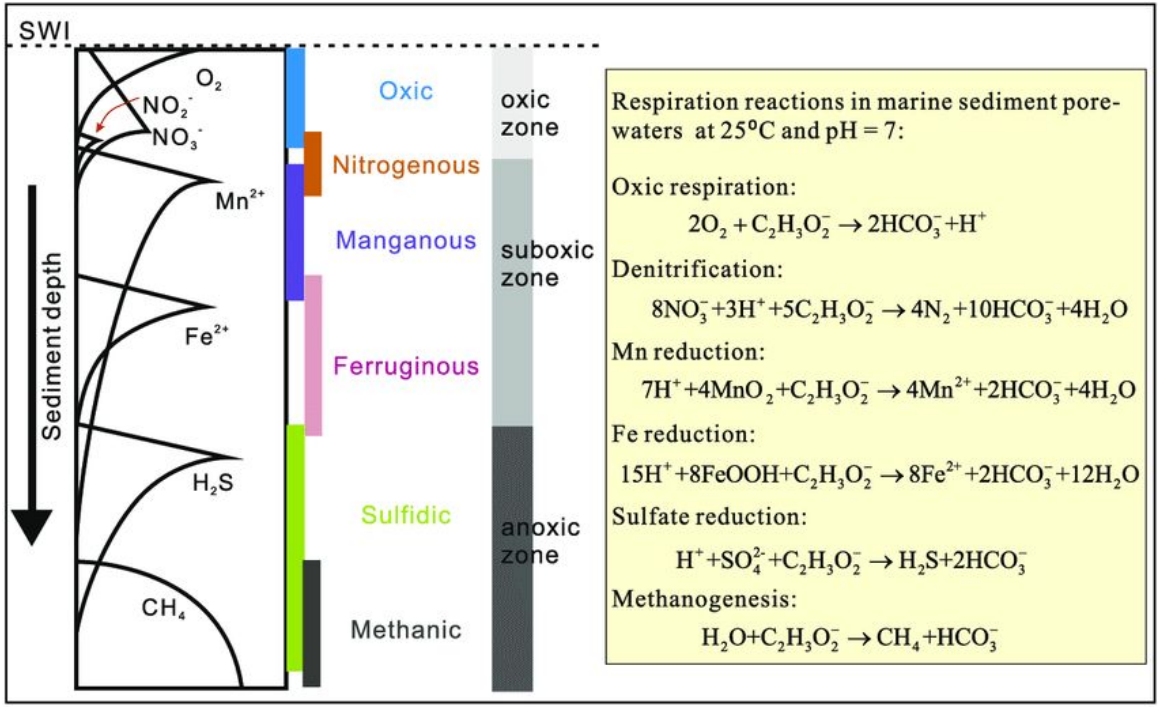


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10.3389/fmicb.2015.00365

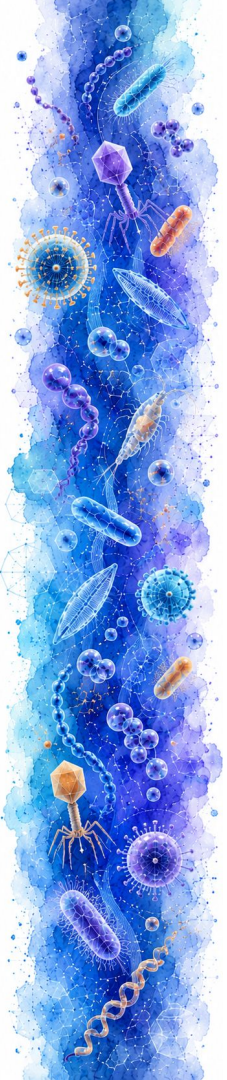
Marine Sediments Microbiome

Lithological and chemical properties in sediments can be subject to alterations related to the degradation of organic matter by microbes during and after deposition. Marine sediments have a **distinct biogeochemical zonation** based on the sequential thermodynamic availability of electron acceptors to oxidize marine planktonic organic matter (organic matter mineralization).



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10.3389/fmicb.2015.00365



This Week Read

Merino N, Aronson HS, Bojanova DP, Feyhl-Buska J, Wong ML, Zhang S and Giovannelli D (2019) Living at the Extremes: Extremophiles and the Limits of Life in a Planetary Context. Front. Microbiol. 10:780. doi: 10.3389/fmicb.2019.00780

Hoshino T, Doi H, Uramoto GI, Wörmer L, Adhikari RR, Xiao N, Morono Y, D'Hondt S, Hinrichs KU, Inagaki F. Global diversity of microbial communities in marine sediment. Proc Natl Acad Sci U S A. 2020 Nov 3;117(44):27587-27597. doi: 10.1073/pnas.1919139117.

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